KING SALMAN INTERNATIONAL UNIVERSITY

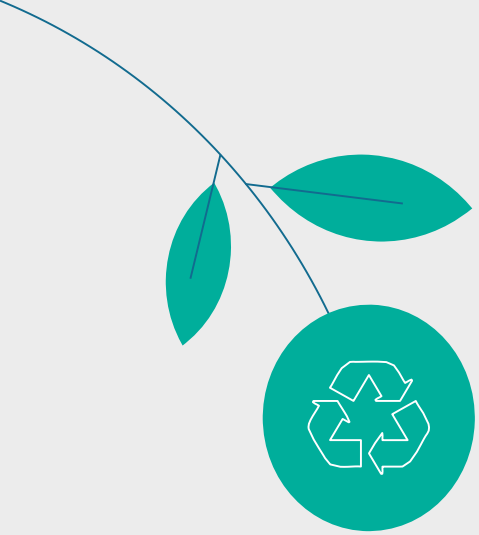
University Requirement

All Programs

Module : Climate Change and Sustainability (ESD 217)

Lectures schedule of Part 1

- Climate Change Concept
- Climate vs. Weather
- Atmosphere layers
- Coral reef bleaching
- Climate change over years
- Climate change causes
- Climate System
- Climate indicators



Citizens in every country need to understand climate change and what it means for them and the world.

The course introduces key climate actions, such as adaptation, low carbon development, climate finance, and climate negotiations. Moreover, it introduces sustainability concept and sustainable development goals

Climate

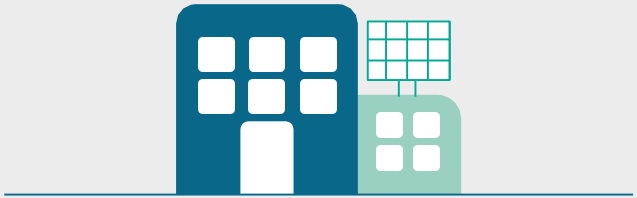
The Climate defined as the "average weather", or as the statistical description in terms of the mean and variability of relevant quantities over a period of time ranging from months to thousands or millions of years.

Weather

The weather is the mix of events that happen each day in our atmosphere. Even though there's only one atmosphere on Earth, the weather isn't the same worldwide. Weather is different in different parts of the world and changes over minutes, hours, days, and weeks

Weather vs. Climate

- *"Weather is one football game, climate is the history of the National Football League."*
- *Climate tells you what clothes to buy, weather tells you what clothes to wear.*



Weather

Refers to short-term changes in the atmosphere



Climate

Describes what the weather is like over a long period of time in a specific area

WEATHER vs CLIMATE

SHORT-TERM STATE OF THE ATMOSPHERE



Rain



LONG-TERM PATTERN OF WEATHER



Tropical Climate



Can change within minutes or hours



Average weather over many years in one specific place

Climate vs. Weather

- *Different regions can have different climates. To describe the Climate of a place, we might say what the temperatures are like during different seasons, how windy it usually is, or how much rain or snow typically falls.*

- *When scientists talk about Climate, they're often looking at averages of precipitation, temperature, humidity, sunshine, wind, and other measures of weather that occur over a long period in a particular place.*

- *In some instances, they might look at these averages over 30 years. And, we refer to these three-decade averages of weather observations as* ***Climate Normal****.*

Climate vs. Weather

❖ *Weather describes what the atmosphere is doing over short time scales; it is what you see when you go outside or look out your window. Weather is extremely variable, changing from hot to cold, rainy to sunny, windy to calm from hour to hour, day to day, or season to season.*

- *Climate is the average weather pattern in a region over long periods of time, typically 30 years or more. For example, California has a Mediterranean climate; that is, historically Californians could expect long, hot, dry summers and cooler, wet winters. The weather changes with the seasons and may vary from year to year (for example, an unusually wet winter), but the general climate pattern is mostly consistent. One way to think about this is that weather is what is happening now, while climate is what you might expect to happen based on many past experiences.*

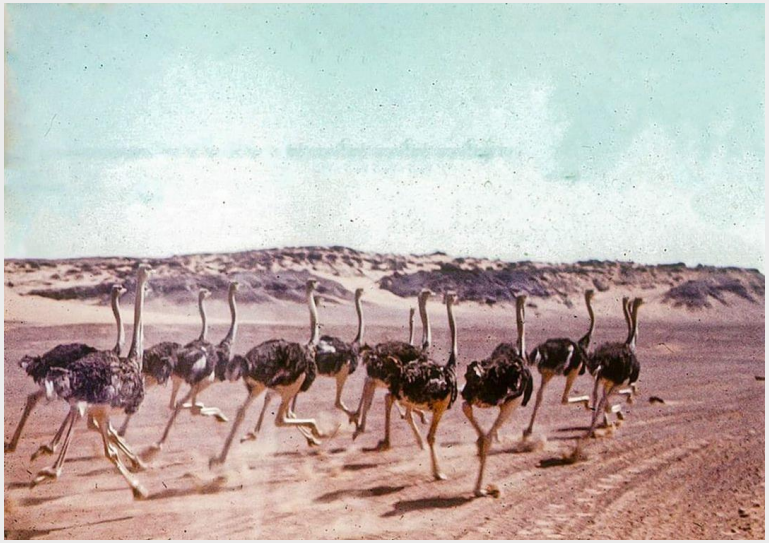
Climate vs. Weather

❖ *Climate, like weather, is variable, only on much longer time scales. Although much of the recent discussion about climate change has centered on global warming, it is important to remember that climate refers not only to temperature, but to precipitation, cloudiness, winds, and relative humidity, among other variables. As we will see, climate has fluctuated on very long-time scales in the past and will continue to do so, due to a variety of natural factors. However, current concerns over climate change relate to how human activities are causing Earth's climate to change, often referred to as anthropogenic climate change or anthropogenic global warming.*

❖ *Anthropogenic refers to the influence that humans can have on the natural world.*

❖ *Average global temperatures have been increasing since before the Industrial Revolution in the 19 century, and they will increase significantly in the future if action is not taken to reduce the emissions of greenhouse gases into the atmosphere. This is because the driving force behind this temperature increase is the increase in concentrations of greenhouse gases in the atmosphere due to human activities. more droughts while others will be subject to stronger storms.*

Examples of Anthropogenic climate change in Egypt





Egyptian Ostrich and Egyptian Ghazal

Anthropogenic climate change “Worldwide”



Polar bear



Seal



Eurasian Wryneck

Climate vs. Weather

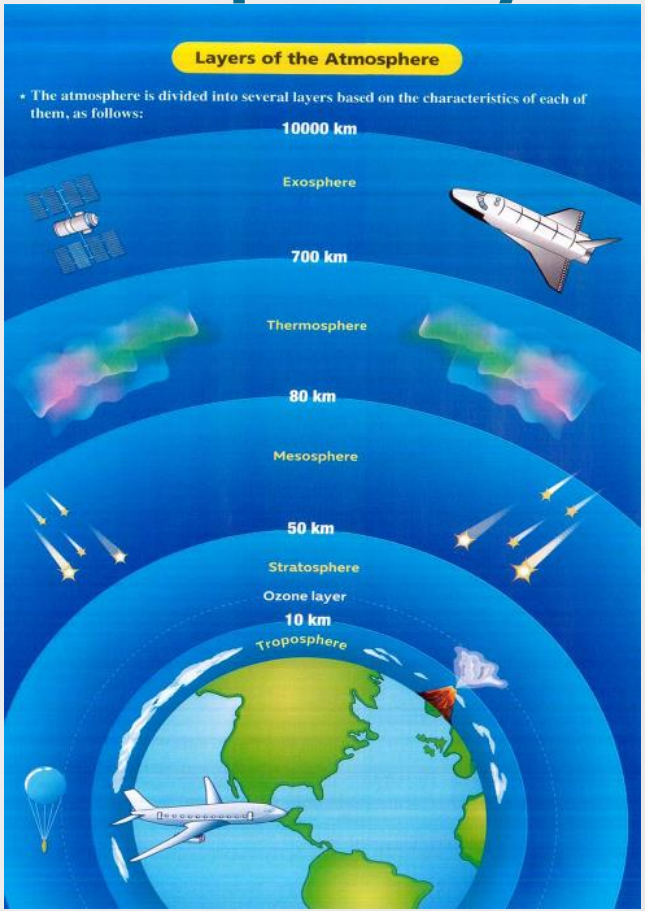
❖ *The term "climate change" is considered more accurate than "global warming," as the climate is not expected to warm uniformly around the globe: Some areas, such as the Arctic, will warm much more than others, while a few might even get cooler.*

❖*Just as important, changes in precipitation will cause some areas to experience more droughts while others will be subject to stronger storms.*

Can you distinguish Weather from Climate?

- It feels hotter than usual right now in Bali (Weather- Climate)
- Freezing temperature in Toronto for 7 days in a row (Weather- Climate)
- The dry season in Kenya is from June to October (Weather- Climate)
- Thailand has high humidity all year around (Weather- Climate)

Atmosphere Layers



Atmosphere Layers
1. Troposphere

Position

It is the closest layer to the Earth's surface in which living organisms live.
The air in this layer is denser compared to other layers.

Importance

It is the layer in which most weather and climate phenomena occur, such as cloud formation, rainfall, wind movement, and others.

Its height above Earth's surface

It extends up to approximately 10 km.

Temperature

The average temperature at Earth's surface is about 15°C.
The air temperature decreases by 6.5°C for every 1000 m (1 km) of elevation in the troposphere.

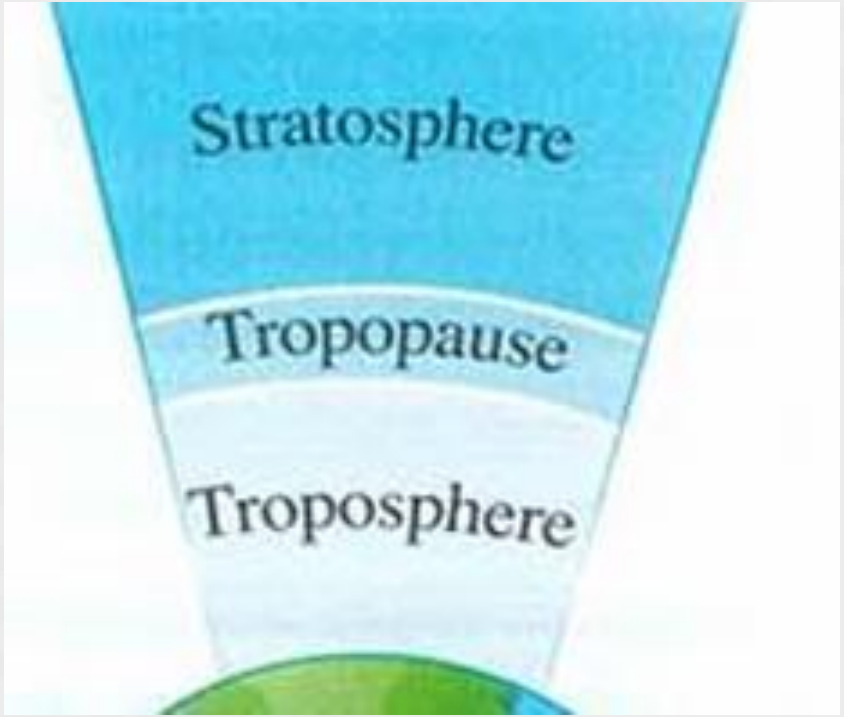
- The essential gases in the troposphere are oxygen, carbon dioxide, water vapor, and other gases provides the fundamental components necessary for life.
- The temperatures are suitable for supporting biological processes.
- The atmospheric pressure is appropriate for most living organisms.

Atmosphere Layers

Tropopause layer

- It is a thin layer of air located at the top of the troposphere, at an altitude of approximately 11 kilometers.

- It acts as a barrier or transitional zone between the troposphere and the stratosphere, where it prevents the mixing of cold air from the troposphere with the warm air in the stratosphere.



Atmosphere Layers
2. Stratosphere

Position

The layer above the troposphere (the second layer from the Earth's surface).

Its height above Earth's surface

It extends from an altitude of 10 km to about 50 km.

Temperature

It does not change through it till an altitude of 20 km from Earth's surface, then the temperature starts to rise as we go higher due to the presence of ozone layer that extends up to about 40 km as ozone gas absorbs the ultraviolet rays (UV) from solar radiation.

Importance

- Contains the ozone layer, that protects the living organisms on Earth from the harmful effects of the short-wave ultraviolet radiation.
- It is considered the preferred layer for aircraft flying since air currents movement in it is horizontal.

Atmosphere Layers
3. Mesosphere

Position

The layer above the Stratosphere (the third layer from the Earth's surface).

Its height above Earth's surface

It extends from an altitude of 50 km to 80 km.

Temperature

It is considered the coldest layer of the atmosphere, with the lowest temperatures that reach −93°C at its top. This is due to the low number of gas molecules capable of absorbing solar radiation.

Importance

Most meteors falling from space burn up and disintegrate after entering the Earth's atmosphere as they pass through this layer, which makes it play a vital role in protecting life on Earth.

Atmosphere Layers
4. Thermosphere

Position

The layer above the Mesosphere (the fourth layer from the Earth's surface).

Its height above Earth's surface

It extends from an altitude of 80 km to 700 km.

Temperature

Temperatures reach higher than 2000°C, where both oxygen and nitrogen gases absorb most of the high frequency solar radiation such as X-rays and gamma rays.

- The lower part of thermosphere, from an altitude of 80 km to 550 km called Ionosphere

Importance

It is used in long-distance radio communications due to its ability to reflect radio waves.

Atmosphere Layers
5. Exosphere

Position

The highest layer in the atmosphere

Its height above Earth's surface

It extends from an altitude of 700 km to 10000 km.

Characteristics

In this layer, some atoms and molecules move over great distances without colliding with other atoms or molecules, which gives them tremendous speeds that may allow them to overcome Earth's gravity and escape into outer space.

Importance

It is considered the ideal region for placing satellites and spacecraft, as they face almost no resistance, which allows them to remain in their orbits for long periods, and this is due to the extremely low air density in this layer.

CORAL BLEACHING

Have you ever wondered how a coral becomes bleached?

HEALTHY CORAL

1 Coral and algae depend on each other to survive.



Corals have a symbiotic relationship with microscopic algae called zooxanthellae that live in their tissues. These algae are the coral's primary food source and give them their color.

STRESSED CORAL

2 If stressed, algae leaves the coral.



When the symbiotic relationship becomes stressed due to increased ocean temperature or pollution, the algae leave the coral's tissue.

BLEACHED CORAL

3 Coral is left bleached and vulnerable.



Without the algae, the coral loses its major source of food, turns white or very pale, and is more susceptible to disease.

WHAT CAUSES CORAL BLEACHING?

Change in ocean temperature
Increased ocean temperature caused by climate change is the leading cause of coral bleaching.

Runoff and pollution
Storm generated precipitation can rapidly dilute ocean water and runoff can carry pollutants — these can bleach near-shore corals.

Overexposure to sunlight
When temperatures are high, high solar irradiance contributes to bleaching in shallow-water corals.

Extreme low tides
Exposure to the air during extreme low tides can cause bleaching in shallow corals.

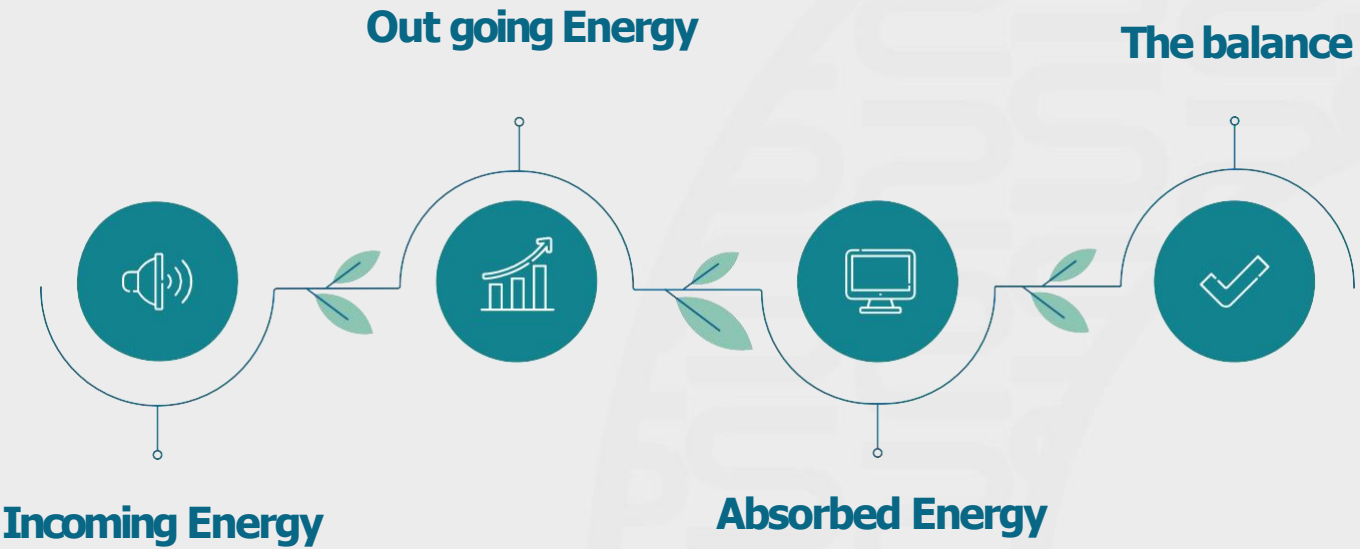
NOAA's Coral Reef Conservation Program
http://coralreef.noaa.gov/





Climate system

The climate system is the result of a delicate energy balance…



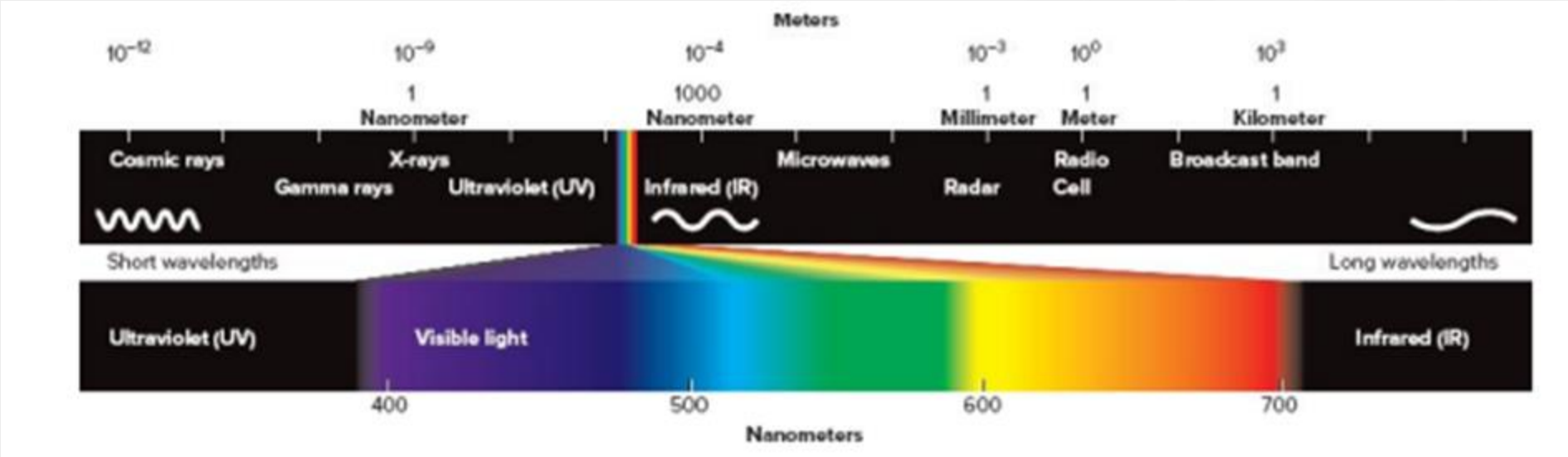
Incoming Energy

- The sun is the single most important source of energy for the earth.
- The temperature on our planet results from the balance between energy coming into the earth from the sun (solar radiation) and the energy leaving the Earth into outer space (Energy balance)
- an estimate of 343 watts per m^2 reaches the planet, but only 240 watts per m^2 passes through the atmosphere
- The Sun is the Earth's primary source of energy. Every second, the Earth absorbs five orders of magnitude (100,000 times) more energy from the Sun than is produced globally from fossil fuel combustion over an entire year.

Incoming Energy

- The Sun transmits its energy to the Earth in the form of electromagnetic radiation. Most of the Sun's radiation and energy are in the visible region of the electromagnetic spectrum, although radiation in both the ultraviolet and infrared regions of the spectrum is also important.
- Electromagnetic radiation is characterized by its wavelength; ultraviolet (UV) radiation has a shorter wavelength than the visible spectrum and infrared (IR) radiation has a longer wavelength.

Incoming Energy

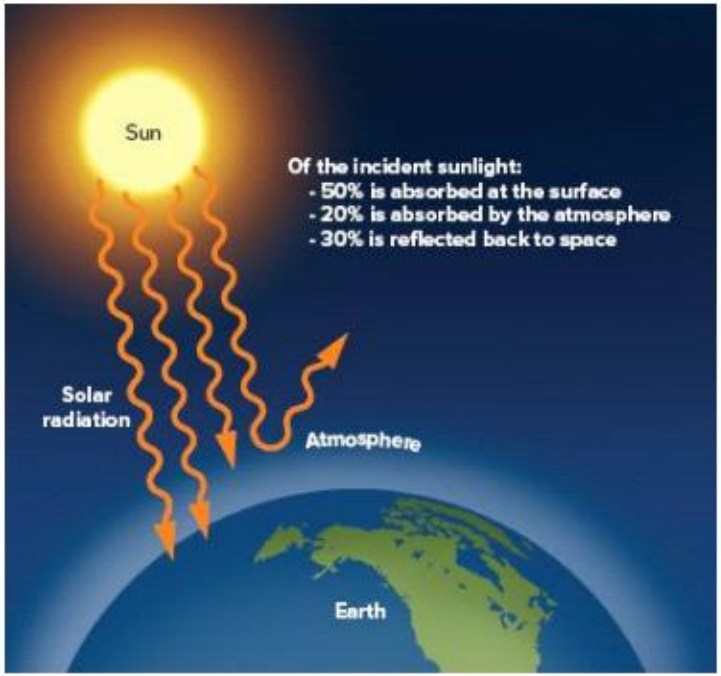


The electromagnetic spectrum

Incoming Energy

- The next figurer shows what happens to the Sun's radiation when it reaches the Earth. About 50% makes its way through the atmosphere and is absorbed by land and sea at the Earth's surface, and 20% is absorbed by the Earth's atmosphere.
- The remaining 30% is reflected back into space by the atmosphere, clouds, and areas of the Earth's surface that are reflective, such as regions covered in snow and ice.

Incoming Energy

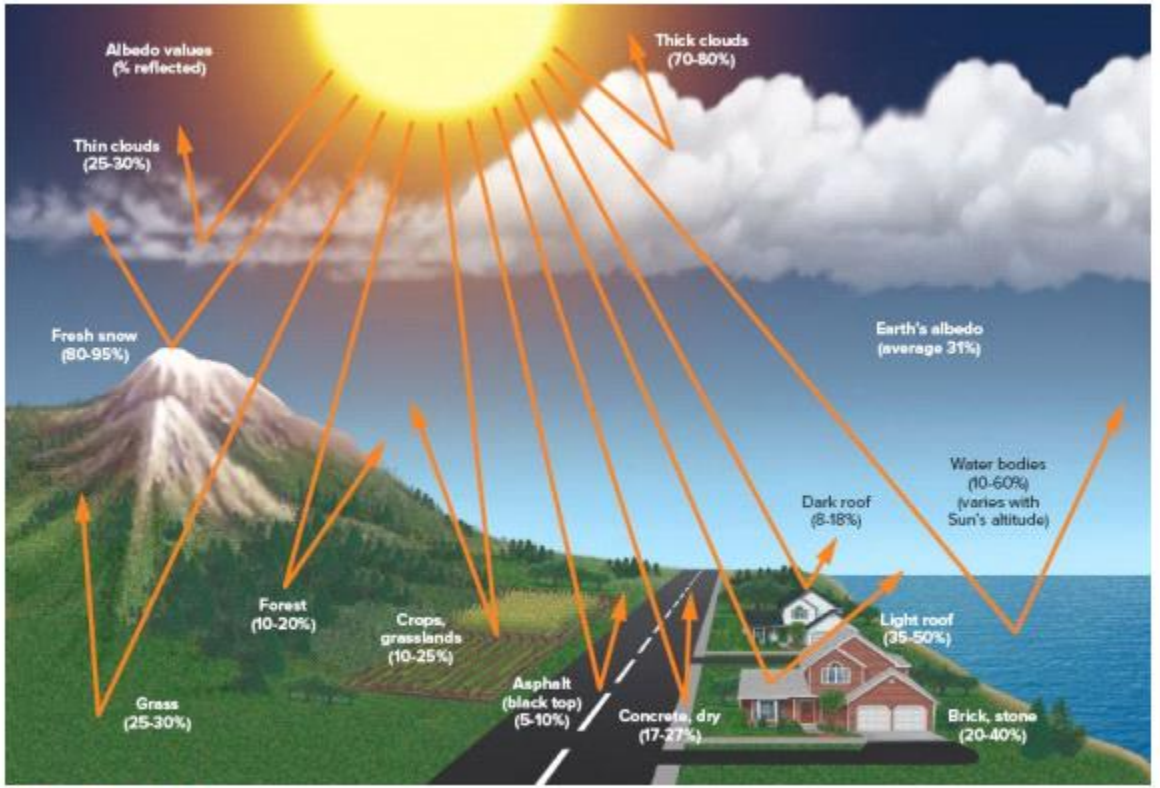


Shortwave solar radiation, illustrated by the orange arrows, is either reflected back to space by clouds, aerosols, gases, or the Earth's surface, absorbed by the atmosphere, or absorbed by the Earth's surface. The absorbed solar radiation is the primary energy input to the Earth's climate system.

Incoming Energy

- The solar radiation that is absorbed at the Earth's surface warms the land and oceans. The percentage of radiation reflected from a surface is referred to as its albedo.
- Lighter-colored objects have a higher albedo than darker ones. The polar regions of the Earth have a much higher albedo than heavily forested areas near the equator. Clouds also have a high albedo; the cloudier the day, the more of the Sun's radiation is reflected back into space. The 30% of solar radiation reflected back into space represents the global average albedo.

Incoming Energy



The albedo of different types of surfaces on the Earth.

Incoming Energy

Black body radiation

- To a reasonable approximation, both the Sun and the Earth can be considered blackbody radiators. That is, the intensity and wavelength of the electromagnetic radiation they emit is dependent on their temperature.
- In fact, all objects give off radiation in a manner that is proportional to their temperature. Hotter objects radiate more energy, and at shorter wavelengths, than cooler objects.

Incoming Energy

Black body radiation

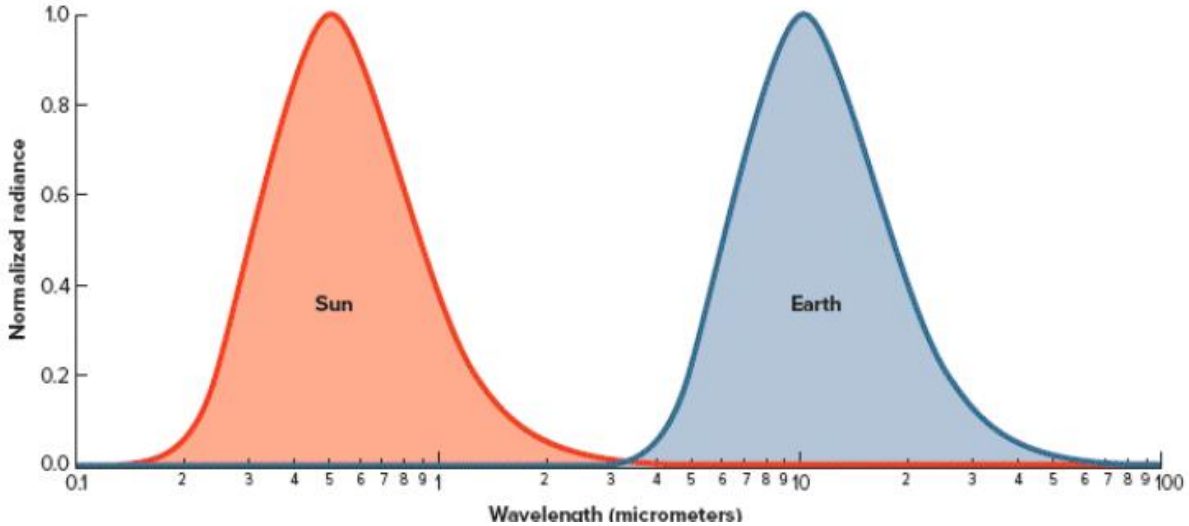
- The radiation emitted by the Sun can be approximated as if the Sun were a blackbody radiator with a temperature of 5,500°C (~10,000°F).
- At this temperature, the peak in the wavelength distribution is in the visible spectrum, around 0.53 micrometers.

Incoming Energy

Black body radiation

- The average surface temperature of the Earth is much cooler, a very comfortable 15°C (59°F). The radiation emitted by the Earth therefore has a longer wavelength, with most being in the infrared region of the electromagnetic spectrum, around 10 micrometers.

Incoming Energy



Incoming Energy

The greenhouse effect

- Greenhouse gases are those that efficiently absorb long-wave radiation. Not all gases in the atmosphere are classified as greenhouse gases. In fact, nitrogen and oxygen, the most abundant atmospheric gases, are not considered greenhouse gases.

Incoming Energy

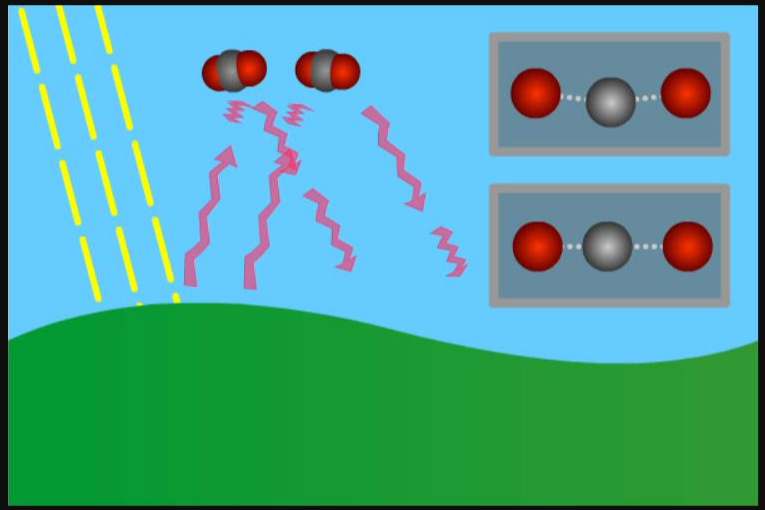
- Our planet, The Earth, is situated in its own greenhouse: the atmosphere. The greenhouse effect is natural, and life on Earth depends on it. Sunlight – energy from the sun – passes through the atmosphere and warms land and water surfaces. The amount of sunlight reaching the Earth is called insolation. Sunlight is perceived as colorless or as “white” light.

Incoming Energy

Did you ever try sunbathing on cliffs on a nice summer's day?
Even after sunset, the warmth radiating from the rocks is noticeable. The sunlight has heated the cliffs throughout the day. The heated cliffs radiate vitality long after dusk. These energy beams are invisible to humans unless special glasses are used to perceive infrared. The military frequently use this. Can we obtain a photograph captured by an infrared camera? The infrared rays of energy released into the atmosphere either escape into outer space or temporarily heat the surrounding air, maintaining warmth even in the absence of sunlight. Unlike other planets in our solar system, the Earth thus has a natural temperature control system. Certain atmospheric gases are critical to this system and are known as greenhouse gases.

Incoming Energy

- On average, about one third of the solar radiation that hits the earth is reflected back to space. Of the remainder, the atmosphere absorbs some, but the land and oceans absorb the most. The Earth's surface becomes warm and as a result emits infrared radiation. Infrared light is heat energy.
- Greenhouse gases are abbreviated GHG.
- The greenhouse gases 'trap' the infrared radiation, thus warming the atmosphere.
- Technically, when the infrared rays hit a GHG molecule, it starts vibrating. When this molecule vibrates, it makes adjacent gas molecules of any kind to vibrate as well. Vibration is a form of energy that is converted to heat energy according to the physical laws of thermodynamics.



Venus versus Earth versus Mars

- The greenhouse effect on Earth raises the average world temperature by around 34°C. How does this compare to the planets nearby?
- Venus is closer to the Sun and gets a lot of sunlight, but it has a reflectiveness (albedo) of 90%. This means it only absorbs 10% of the sunlight that hits it. So, Venus has a stable temperature that is very cold, at about -89°C. But the actual temperature on Venus is about 477°C!
- Venus is extremely hot not because it's near the Sun, but because it has a strong greenhouse effect.

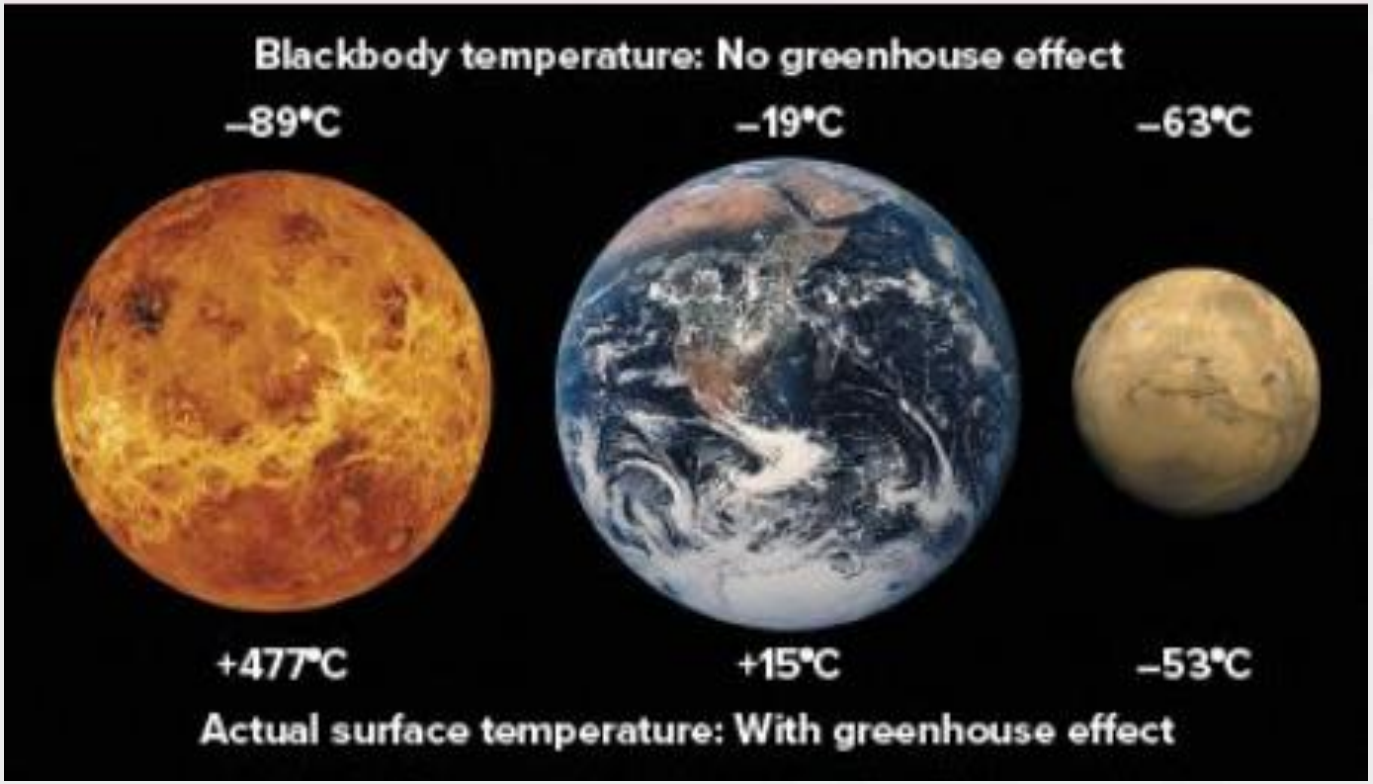
Venus versus Earth versus Mars

- The atmosphere of Venus is 96.5% CO_2 compared to just 0.039% in the Earth's atmosphere. Also, the atmospheric pressure on Venus is ninety-two times greater than that on the Earth, which means the amount of CO_2 is extremely high.
- Mars has an albedo of 0.25, similar to the Earth. However, because Mars is farther away from the Sun it receives less sunlight, and its blackbody temperature is only −63°C.

Venus versus Earth versus Mars

- Similar to Venus, the Martian atmosphere is 95.3% CO_2. However, the Martian atmosphere has a pressure that is only 1/1,000th that of the Earth. Because of the low atmospheric pressure, the greenhouse effect on Mars is very weak, and the mean surface temperature is very similar to the blackbody temperature.
- Thus, by examining our neighboring planets, we can see that the greenhouse effect is very sensitive to concentrations of greenhouse gases in the atmosphere.

Venus versus Earth versus Mars



Outgoing energy

- About half the solar radiation striking the Earth and its atmosphere is absorbed in the surface.
- The other half is absorbed by the atmosphere or reflected back into space by clouds, small particles in the atmosphere, snow, ice and deserts at the earth's surfaces.
- Part of the energy absorbed at the earth's surface is radiated back or re admitted to the atmosphere and space in the form of heat or thermal energy.

Outgoing energy

Incoming energy		Outgoing energy	
Units	Source	Units	Source
+100	Shortwave radiation from the sun.	-23	Shortwave radiation reflected back to space by clouds.
		-7	Shortwave radiation reflected to space by the earth's surface.
		-49	Longwave radiation from the atmosphere into space.
		-9	Longwave radiation from clouds into space.
		-12	Longwave radiation from the earth's surface into space.
+100	Total Incoming	-100	Total Outgoing

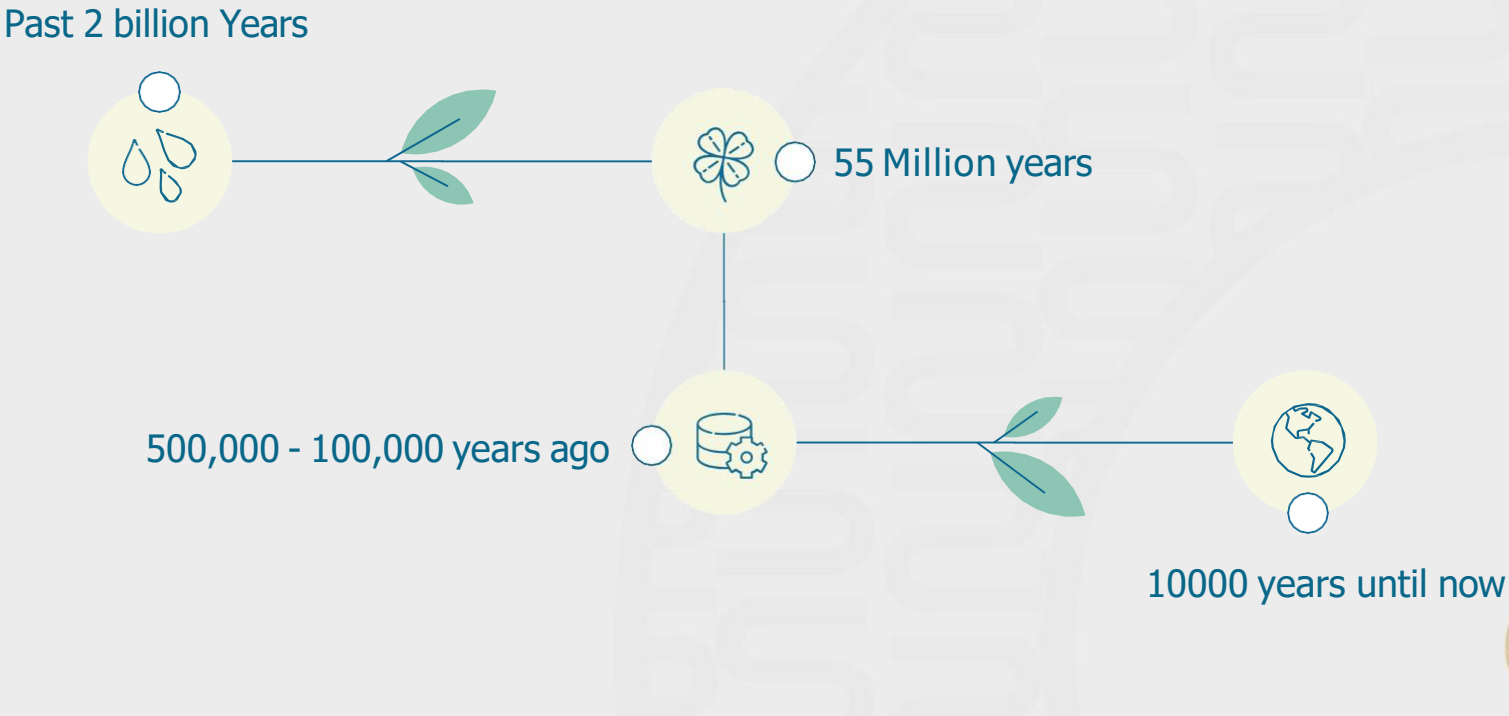
Absorbed energy

- Solar energy is absorbed by the Earth's surface and warms it.
- The conversion of energy into heat causes the emission of long-wave (infrared) radiation back into the atmosphere
- Not all radiation makes it out into space –some is trapped by clouds and greenhouse gases and is reemitted back to the Earth's surface

The balance

- The temperature we feel is the balance of the heat energy that is trapped on Earth by the planet's atmosphere.
- The thermal radiation absorbed and reflected back to the Earth's surface by Greenhouse gas allows for an average temperature on Earth of around 14°c well above 19 °c
- More GHG in the atmosphere can trap more heat, and lead to a gradual increase in temperature and changes in the climate

How has the climate on Earth evolved in the past?



Climate change over years

- Over long-time scales, the Earth's climate is determined by a few primary factors: the solar input, the concentration of greenhouse gases, and the average global albedo.
- All of these have varied dramatically over time.
- In order to understand how climate depends on these factors, and thus be able to predict how climate is likely to change in the near and long-term future, we need to look back in time at ancient climate, a field of study known as **paleoclimatology**.
- Paleoclimate studies require that scientists find paleoclimate proxies—materials that have retained a unique signature of the ancient climate conditions.

Climate change over years

- The signatures that can be captured are not limited to temperature. Proxies have been found that provide information on ice volume, sea level, wind speed, precipitation, and greenhouse gas levels.
- These proxies can take many forms. A few examples among many are the number of pollen spores per volume of sediment; the width of tree rings; the concentrations of noble gases in groundwaters; the stable isotopic composition of many different items (e.g., ice cores, carbonate minerals, fossils); and the composition of air bubbles trapped in ice cores.

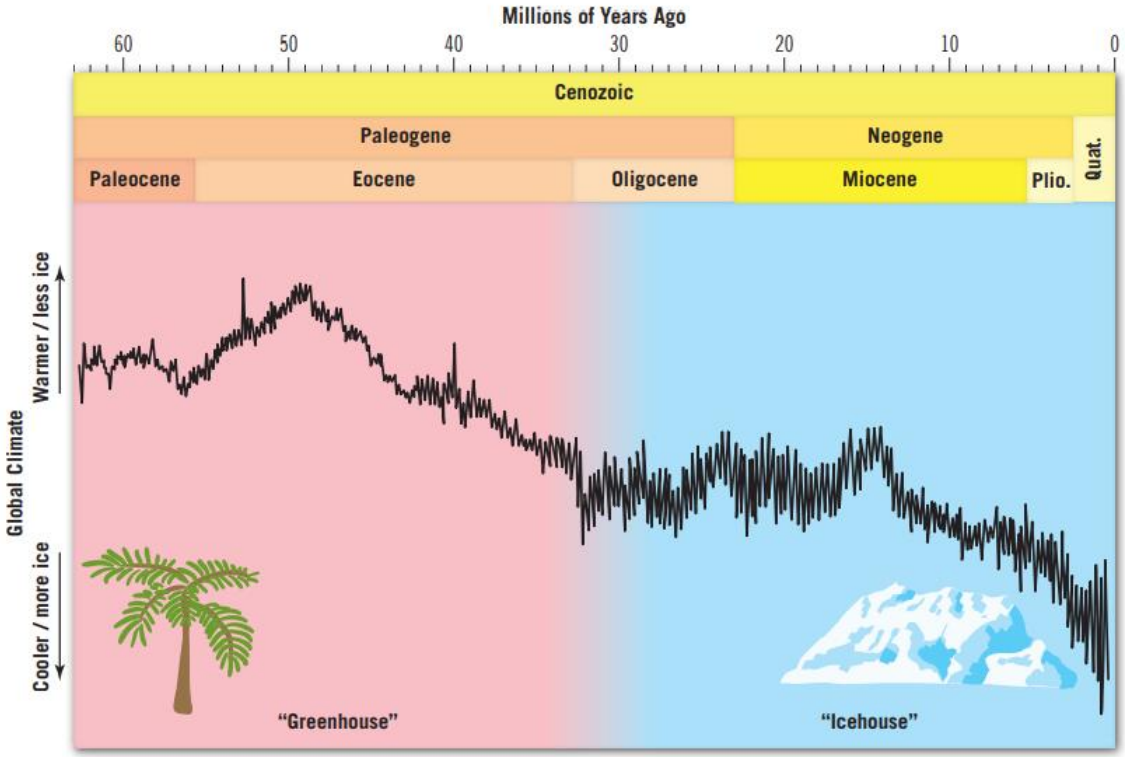
Climate change over years

- We must take into consideration changes in the continental geography, variations in solar input, and variations in greenhouse gas concentrations over very long time periods. Prior to 2.6 million years ago, during pre-Quaternary times, global temperatures were generally higher than they are now. These higher temperatures were typically correlated with atmospheric CO_2 levels that were much greater than they are today.
- For instance, temperatures were higher and atmospheric CO_2 concentrations were above 1,000 parts per million (ppm) during the Early Eocene (52–48 million years ago), which is more than three times as high as in modern preindustrial periods.

Climate change over years

- And during the Paleocene–Eocene Thermal Maximum (PETM), a brief period at the boundary between the Paleocene and Eocene lasting only a few tens of thousands of years, there was a massive release of carbon into the atmosphere that precipitated a large increase in global temperatures; the origins of this massive carbon release are still debated by scientists.
- This indicates that there is a connection between higher greenhouse gas levels and warmer climates.

Climate change over years



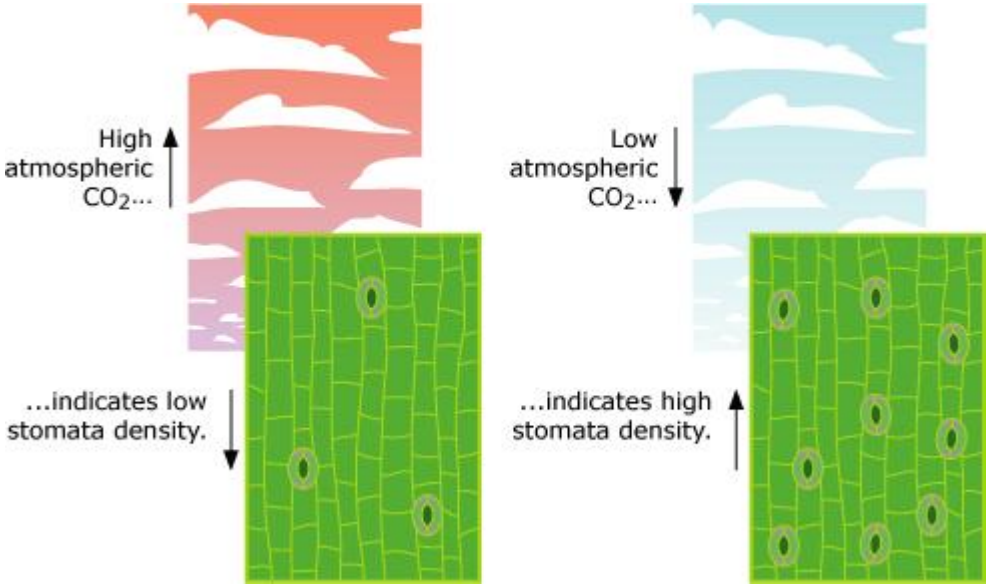
Climate change over years

- How do we know about these paleoclimatic changes?

1. Stomata in ancient leaves
2. Lush forests and grasslands
3. Seafloor sediment: A storehouse of climate Data
4. Oxygen Isotope Analysis
5. Glacial Ice
6. Tree rings
7. Other types of proxy Data “ pollen, corals, historical documents”

Climate change over years :Stomata

- The isotopic composition of carbon in fossils and the density of stomata in ancient leaves can be used to recreate atmospheric CO2 concentrations from millions of years ago. In plants, stomata regulate the flow of CO_2 and H_2O into and out of the plant.



Climate change over years: Lush and grassland

- Even at high latitudes, relatively moderate temperate climates prevail during greenhouse times, when there is little to no permanent ice at either pole. The global climate is sufficiently cool to sustain ice sheets at one or both poles during icehouse conditions.
- Only a few times in the last 541 million years, or the Phanerozoic (or "visible life") eon, has Earth's temperature shifted gradually between these two categories ice house and hot or green house. Numerous fossils found in Phanerozoic rocks and sediments record significant environmental and evolutionary processes. During the Cenozoic era, the most recent change took place.

Climate change over years: Lush and grassland

- The early Cenozoic was a time of greenhouse climates like those the dinosaurs experienced during the preceding Mesozoic era.
- By about 34 million years ago, permanent ice sheets were present at the South Pole, ushering in icehouse conditions. Climate warmed during the Miocene epoch (about 20 million years ago) as mammal populations reached their greatest diversity.
- Climate then cooled. In North America, the lush “greenhouse” forests (there were palm trees in Wyoming and banana trees in Oregon) were replaced by open grasslands.

Climate change over years: Lush and grassland

- An "icehouse" climate that is colder and drier is more suitable for grassland ecosystems.
- The Earth's climate was sufficiently frigid to sustain massive ice sheets at both poles by 2 million years ago, marking the beginning of the Quaternary epoch. Ice moved almost as far south in the Northern Hemisphere as the modern-day Ohio River before retreating to Greenland.
- This cycle of ice advance and retreat has been happening roughly every 100,000 years for the last 800,000 years. It was approximately 18,000 years ago that the last significant ice sheet advance peaked.

Climate change over years



Sea floor sediment: A storehouse of climate data

- We are aware that the Earth system's components are interconnected, meaning that modifications to one can have an impact on all or some of the others. So, the nature of marine life is affected by variations in oceanic and atmospheric temperatures.
- The remains of species that formerly resided close to the sea surface (the ocean–atmosphere interface) can be found in the majority of seafloor sediments. The shells of these near-surface animals gradually sink to the ocean floor after they die, becoming a part of the sedimentary record.

Sea floor sediment: A storehouse of climate data

- These seafloor sediments are useful recorders of worldwide climate change because the numbers and types of organisms living near the sea surface change with the climate.
- For this reason, scientists are tapping the huge reservoir of data in seafloor sediments. The sediment cores gathered by drilling ships and other research vessels have provided invaluable data that have significantly expanded our knowledge and understanding of past climates.

oxygen Isotope Analysis

- One significant source of proxy data on historical climate conditions is the oxygen isotopes found in water molecules or in the shells of marine species. The exact measurement of the ratio in the shells of certain bacteria buried in deep-sea sediments is the foundation of oxygen isotope analysis.
- A period when ice sheets were expanding is indicated by a greater ratio of $^{18}O/^{16}O$ in shells.
- Temperature also affects the $^{18}O/^{16}O$ ratio.
- When temperatures are high, more ^{18}O evaporates from the oceans, and when temperatures are low, less. As a result, the heavy isotope is less prevalent during colder times and more prevalent during warm eras' precipitation. Scientists have used this idea to determine historical temperature variations by examining the layers of snow and ice in glaciers.

climate change recorded in Glacial Ice

- Data from ice cores is crucial for recreating historical climates by using vertical cores extracted from the Antarctic and Greenland ice sheets.
- Using a drilling rig, which is essentially a miniature oil drill, scientists gather samples. The drill head is followed into the ice by a hollow shaft, which extracts the ice core.
- This allows for the acquisition of cores for analysis, which can be up to 2000 meters (6500 feet) long and may include over 200,000 years of climate history.
- By using oxygen isotope analysis, scientists may create a record of fluctuating air temperatures and snowfall.
- Changes in the composition of the atmosphere are also recorded by air bubbles trapped in the ice. Temperature variations are associated with changes in carbon dioxide and methane. Additionally, the cores contain atmospheric fallout from volcanic ash, wind-blown dust, pollen, and contemporary pollution.

tree rings: Archives of environmental History

- Tree rings provide valuable information about historical climates, as they reveal the thickness and density of tree ring, which indicate the environmental conditions during the year. Tree-ring patterns are similar for trees growing simultaneously in the same area. The age of a tree can be determined by counting growth rings annually and obtaining nondestructive core samples from living trees.



Tree rings **A.** Each year a growing tree produces a layer of new cells beneath the bark. If the tree is cut down and the trunk is examined, each year's growth can be seen as a ring. These rings are useful records of past climate because the amount of growth (the thickness of a ring) depends on precipitation and temperature. (Photo by Victor Zastolskiy/Fotolia) **B.** Scientists are not limited to working with trees that have been cut down. Small, nondestructive core samples can be taken from living trees. (Photo by Gregory K. Scott/Science Source)

Other types of proxy Data

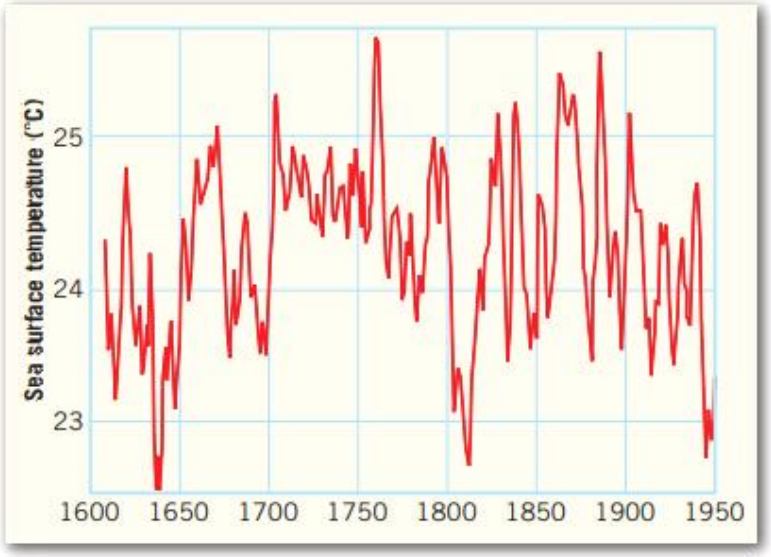
- In addition to the sources already discussed, other sources of proxy data that are used to gain insight into past climates include fossil pollen, corals, and historical documents.
- **Fossil pollen:** Climate is a major factor influencing the distribution of vegetation, so the nature of the plant community occupying an area is a reflection of the climate. Pollen and spores are parts of the life cycles of many plants, and because they have very resistant walls, they are often the most abundant, easily identifiable, and best-preserved plant remains in sediments. By analyzing pollen from accurately dated sediments, scientists can obtain high-resolution records of vegetation changes in an area. Past climates can be reconstructed from such information.

Other types of proxy Data

- **Corals:** Coral reefs consist of colonies of corals, invertebrates that live in warm, shallow waters and form atop the hard material left behind by past corals. Corals build their hard skeletons from calcium carbonate ($CaCO_3$) extracted from seawater. The carbonate contains isotopes of oxygen that can be used to determine the temperature of the water in which the coral grew. The portion of the skeleton that forms in winter has a different density than the portion that forms in summer because of variations in growth rates related to temperature and other environmental factors. Thus, corals exhibit seasonal growth bands very much like those observed in trees. The accuracy and reliability of the climate data extracted from corals has been established by comparing recent instrumental records to coral records for the same period.

Other types of proxy Data

Corals record sea-surface temperatures Coral colonies thrive in warm, shallow tropical waters. The tiny invertebrates extract calcium carbonate from seawater to build hard parts. They live atop the solid foundation left by past coral. Chemical analysis of the changing composition of coral reefs with depth can provide useful data on past near-surface temperatures. This graph shows a 350-year record of sea-surface temperatures obtained through oxygen isotope analysis of coral from the Galapagos Islands.



isotope analysis of coral-growth rings can also serve as a proxy measurement for precipitation, particularly in areas where large variations in annual rainfall occur.

Think of coral as a *paleothermometer* that enables us

Other types of proxy Data

- **Corals:** Oxygen isotope analysis of coral-growth rings can also serve as a proxy measurement for precipitation, particularly in areas where large variations in annual rainfall occur. Think of coral as a paleothermometer that enables us to answer important questions about climate variability in the world's oceans.
- **Historical Documents** it sometimes contain helpful information. Although it may seem that such records should readily lend themselves to climate analysis, that is not the case. Most manuscripts were written for purposes other than climate description. Furthermore, writers understandably neglected periods of relatively stable atmospheric conditions and mention only droughts, severe storms, memorable blizzards, and other extremes. Nevertheless, records of crops, floods, and human migration have furnished useful evidence of the possible influences of changing climate

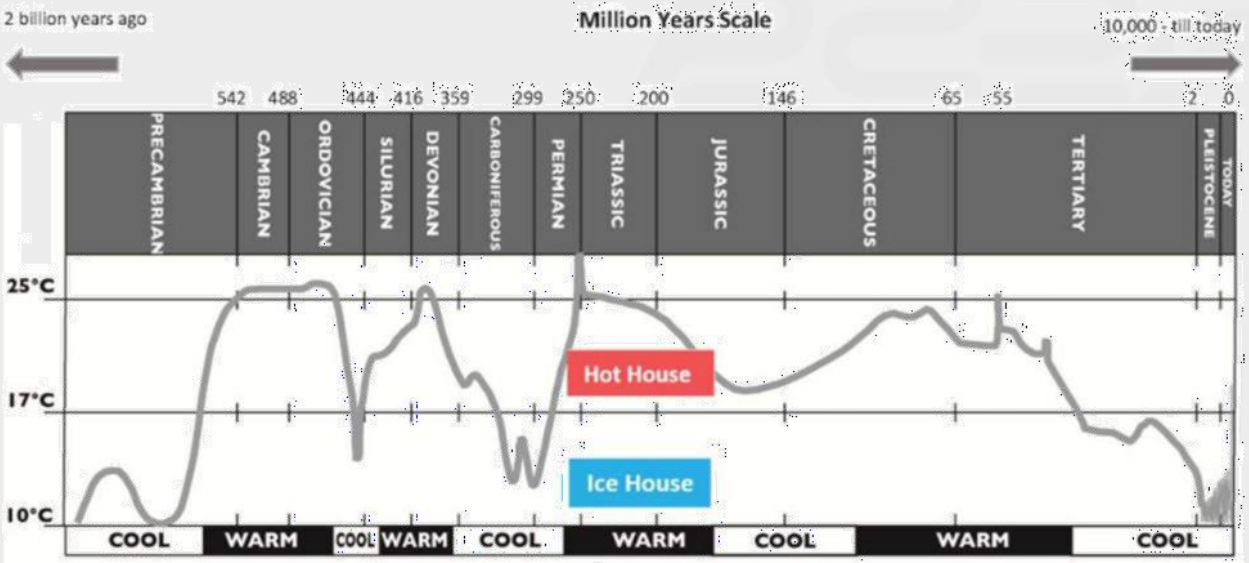
other types of proxy Data



Harvest dates as climate clues Historical records can sometimes be helpful in the analysis of past climates. The date for the beginning of the grape harvest in the fall is an integrated measure of temperature and precipitation during the growing season. These dates have been recorded for centuries in Europe and provide a useful record of year-to-year climate variations. (Photo by SGM/AGE Fotostock)

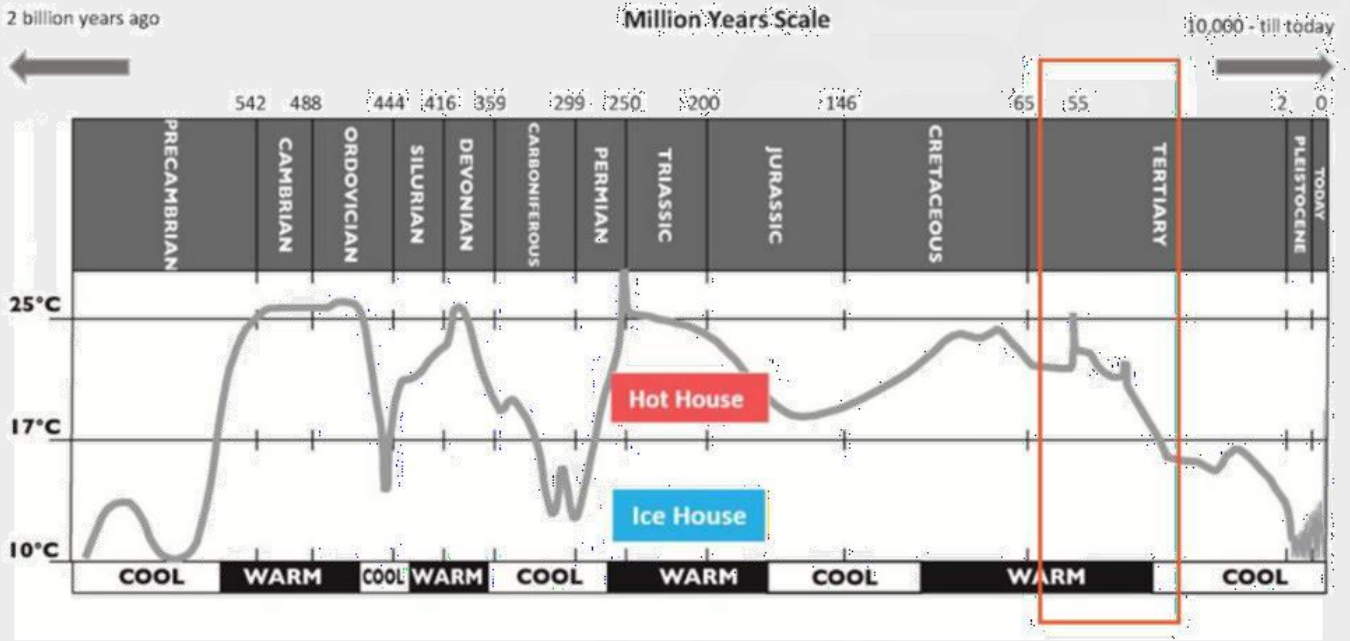
Past 2 billion Years

During the last 2 billion years the Earth's climate has alternated between a frigid "Ice House", and a steaming “Hothouse", like the world of the dinosaurs. During the ‘greenhouse’ period, there is a lot more liquid water on the planet, while during ‘icehouse’ world the climate is cold enough to support large sheets of ice, sometimes covering the entire planet



55 Million years

The most recent transition between those phases occurred about 55 million years ago when the temperature reached a thermal maximum followed by a long period of cooling, which we are currently experiencing.



500,000 - 100,000 years ago

500,000 - 100,000 years ago a period of hot and cold fluctuations occurred. Samples from the Arctic ice show that over a period of several hundred thousand years, the great ice sheets that covered parts of North America and Europe have melted in a series of temperature surges, each one occurring approximately every 100,000 years. This leads us to the last 10,000 years.

10000 years until now

The last 10,000 years are known as the Holocene. They have been uniquely stable and warm. Sea level stabilized at its current level about 7,000 years ago. This has allowed human civilization to develop. Even though temperatures were fairly stable compared to previous periods, there have still been slight climate variations in the recent climate history, such as a temperature decline between about 1600 and 1850, known as the Little Ice Age.

What causes changes in climate?

Earth’s climate does not change without a reason. Many factors may influence it over long periods of time. Such factors are known as "climate forcing".



Solar variability



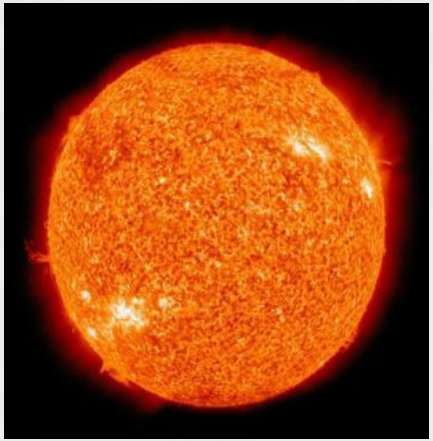
Volcanic activities



Changes in the Carbon cycle

Solar variability

Solar variation, the change in the amount of radiation emitted by the Sun, or small changes in the Earth's rotation in orbit may change the amount of energy Received by the sun. However, in the past 140 years, we have experienced very little solar variation



Solar variability

- Among the most persistent hypothesis of climate change have been those based on the idea that the Sun is a variable star and that its output of energy varies through time.
- The effect of such changes would seem direct and easily understood: Increases in solar output would cause the atmosphere to warm, and reductions would result in cooling.
- This idea is appealing because it can be used to explain climate change of any length or intensity.

Solar variability

- However, no major long-term variations in the total intensity of solar radiation have yet been measured outside the atmosphere.
- Such measurements were not even possible until satellite technology became available.
- We can now measure solar output but still need many decades of records before we will begin to sense how variable (or invariable) energy from the Sun really is.

Solar variability

- Some hypotheses for climate change have related to sunspot cycles. The most conspicuous and best-known features on the surface of the Sun are the dark blemishes called sunspots.
- Sunspots are huge magnetic storms that extend from the Sun's surface deep into the interior. Moreover, these spots are associated with the Sun's ejection of huge masses of particles that, on reaching Earth's upper atmosphere, interact with gases to produce displays known as the aurora borealis, or Northern Lights, in the Northern Hemisphere.

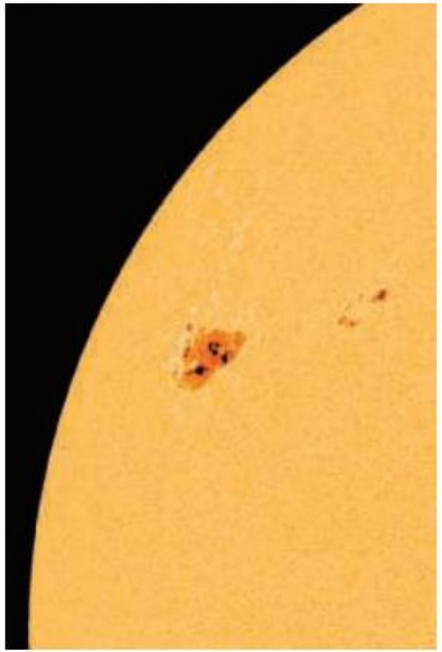
Solar variability



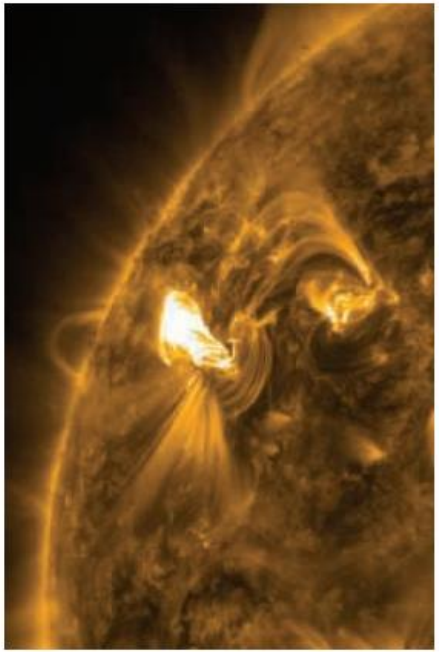
aurora borealis

Solar variability

This view shows an approximation of the Sun's surface. The black spots surrounded by deep orange is a sunspot region where magnetic activity is extremely intense.



The instrument that produced this image used ultraviolet, radio, and other parts of the electromagnetic spectrum. Looping lines show solar plasma following magnetic field lines.



Solar variability

- Sunspots occur in cycles, with the number of sunspots reaching a maximum about every 11 years. During periods of maximum sunspot activity, the Sun emits slightly more energy than during sunspot minimums.
- Based on measurements from space that began in 1978, the variation during an 11-year cycle is about 0.1 %.
- Although sunspots are dark, they are surrounded by brighter areas, which apparently offset the effect of the dark spots. It appears that this change in solar output is too small, and the cycles are too short to have any appreciable effect on global temperatures. However, there is a possibility that longer-term variations in solar output may affect climates on Earth.

Solar variability

- For example, the span between 1645 and 1715 is a period known as the Maunder minimum, during which sunspots were largely absent. This period of missing sunspots closely corresponds with a period in climate history known as the Little Ice Age, an especially cold period in Europe.
- For some scientists, this correlation suggests that a reduction in the Sun's output was likely responsible at least in part for this cold episode. Other scientists seriously question this notion.
- Their hesitation stems in part from subsequent investigations using different climate records from around the world that failed to find a significant correlation between sunspot activity and climate.

Volcanic activities

Unpredictable volcanic eruptions can also cause climate disruption. Large eruption inject aerosols into the atmosphere, which cools the earth's surface.



Volcanic activity

- The idea that explosive volcanic eruptions might alter Earth's climate was first proposed many years ago. It is still regarded as a believable explanation for some aspects of climatic variability. Explosive eruptions emit huge quantities of gases and fine-grained debris into the atmosphere.

- The greatest eruptions are sufficiently powerful to inject material high into the atmosphere; strong upper-level winds then spread it around the globe, where it remains for many months or even years.

Volcanic activity

This satellite image shows the sulfur dioxide (SO_2) plume in shades of purple and black. Climate may be affected when large quantities of SO_2 are injected into the atmosphere.

This image was taken from the International Space Station and shows a plume of volcanic ash streaming southeastward from the volcano.

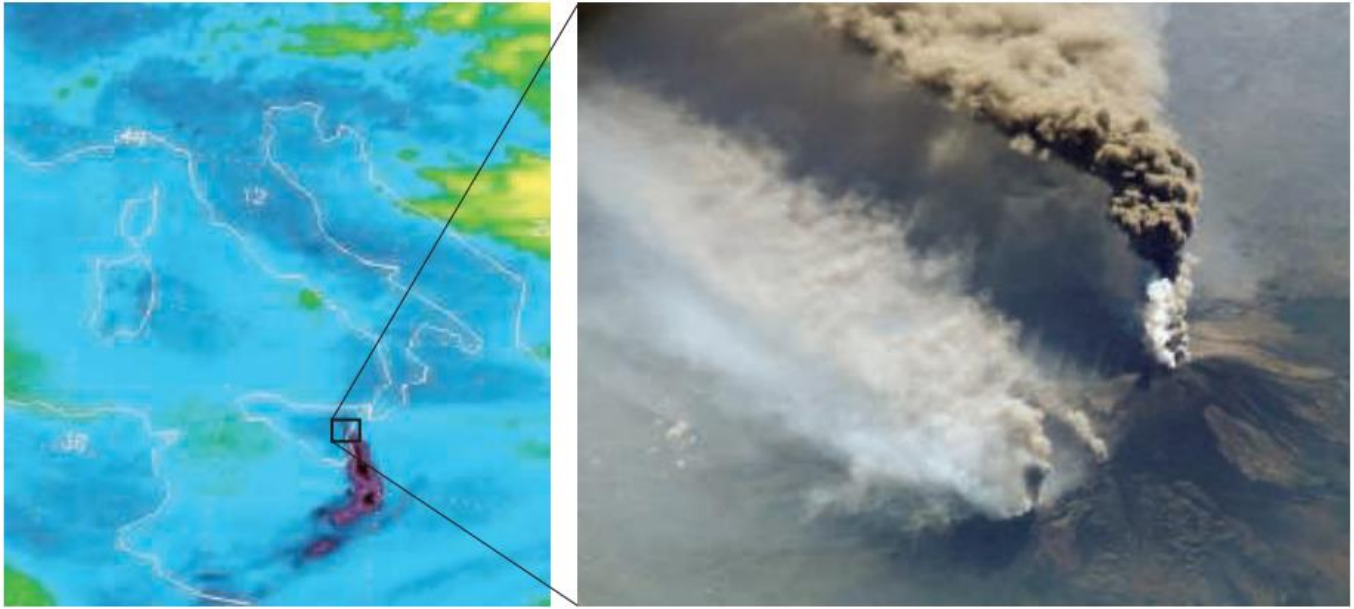


Figure 21.19
Mount Etna erupting in October 2002 This volcano on the island of Sicily is Europe's largest and most active volcano. (NASA images)

Volcanic activity “the effect of Volcanic Aerosols on climate”

- Suspended volcanic material filters out a portion of the incoming solar radiation, which in turn lowers temperatures in the troposphere. More than 200 years ago, Benjamin Franklin used this idea to argue that material from the eruption of a large Icelandic volcano could have reflected sunlight back to space and therefore might have been responsible for the unusually cold winter of 1783–1784.
- Perhaps the most notable cool period linked to a volcanic event is the “year without a summer” that followed the 1815 eruption of Mount Tambora in Indonesia. Tambora's is the largest eruption of modern times. During April 7–12, 1815, this nearly 4000m high volcano violently expelled an estimated 100 Km^3 of volcanic debris. The impact of the volcanic aerosols on climate is believed to have been widespread in the Northern Hemisphere.

Volcanic activity “the effect of Volcanic Aerosols on climate”

- From May through September 1816, a record series of cold spells affected the northeastern United States and adjacent portions of Canada. There was heavy snow in June and frost in July and August. Abnormal cold was also experienced in much of Western Europe. Similar, although apparently less dramatic, effects have been associated with other great explosive volcanoes, including Indonesia’s Krakatoa in 1883
- Three more recent volcanic events have provided considerable data and insight regarding the impact of volcanoes on global temperatures. The eruptions of Washington State’s Mount St. Helens in 1980, the Mexican volcano El Chichón in 1982, and Mount Pinatubo in the Philippines in 1991 have given scientists opportunities to study the atmospheric effects of volcanic eruptions with the aid of more sophisticated technology than was previously available.

Volcanic activity “Volcanism and Global Warming”

- Satellite images and remote-sensing instruments allowed scientists to closely monitor the effects of the clouds of gases and ash that these volcanoes emitted.
- The Cretaceous period is the last period of the Mesozoic era, the era of middle life that is often called the “age of dinosaurs.” It began about 145.5 million years ago and ended about 65.5 million years ago, with the extinction of the dinosaurs (and many other life-forms as well).
- The Cretaceous climate was among the warmest in Earth’s long history. Dinosaurs, which are associated with mild temperatures, ranged north of the Arctic Circle. Tropical forests existed in Greenland and Antarctica, and coral reefs grew as much as 15° latitude closer to the poles than at present.

Volcanic activity “Volcanism and Global Warming”

- Deposits of peat that would eventually form widespread coal beds accumulated at high latitudes. Sea level was as much as 200 meters (650 feet) higher than it is today, consistent with a lack of polar ice sheets.
- What caused the unusually warm climates of the Cretaceous period? Among the significant factors that may have contributed was an enhanced greenhouse effect due to an increase in the amount of carbon dioxide in the atmosphere. But where did the additional CO_2 come from?
- Many Scientists suggest that the probable source was volcanic activity. Carbon dioxide is one of the gases emitted during volcanism, and there is now considerable geologic evidence that the Middle Cretaceous was a time when there was an unusually high rate of volcanic activity.

Volcanic activity “Volcanism and Global Warming”

- Several huge oceanic lava plateaus were produced on the floor of the western Pacific during this span.
- These vast features were associated with hot spots that may have been produced by large mantle plumes. Massive outpourings of lava over millions of years would have been accompanied by the release of huge quantities of CO_2, which in turn would have enhanced the atmospheric greenhouse effect. Thus, the warmth that characterized the Cretaceous may have had its origins deep in Earth’s mantle.

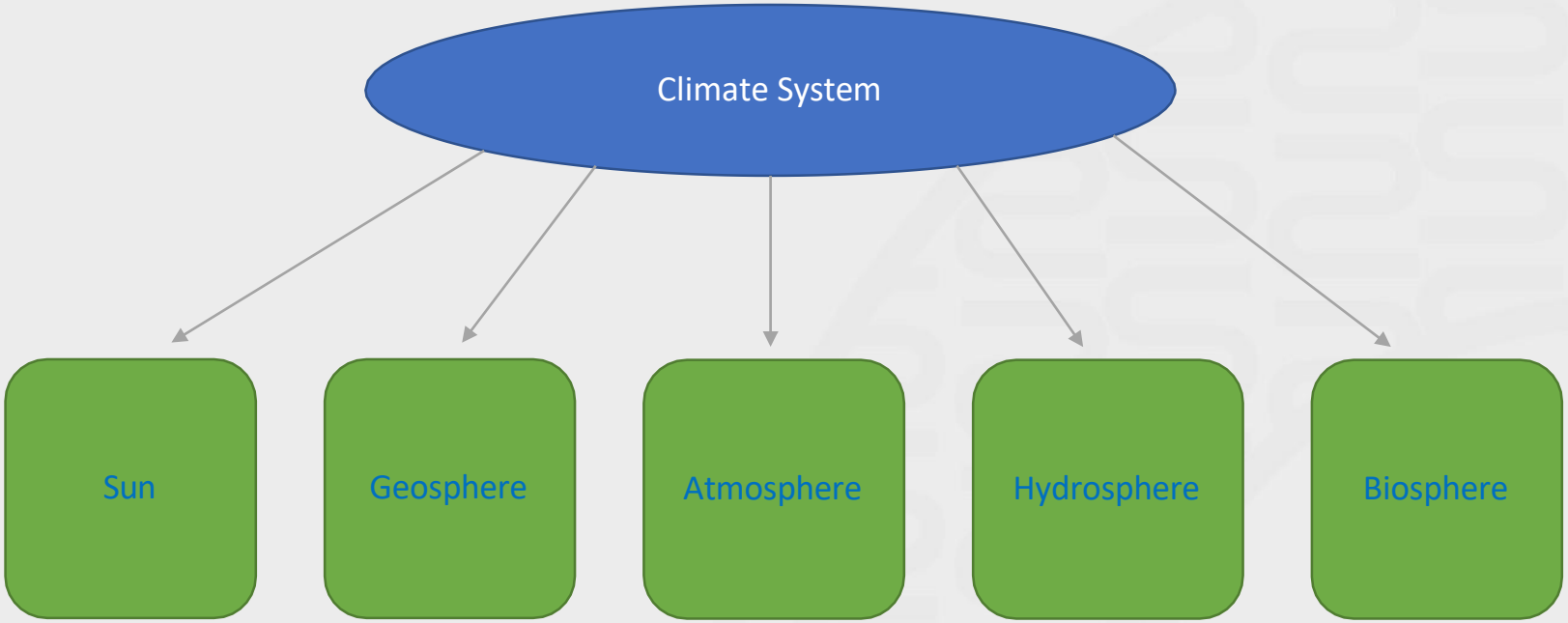
Changes in the Carbon cycle

When is the amount of carbon dioxide and other Greenhouse gases in the atmosphere increases, this has a direct casual effect on changes in the climate.



Climate system




Climate System
Sun
Geosphere
Atmosphere
Hydrosphere
Biosphere

01. Sun

Sun is the source of an enormous amount of energy, a portion of which provides Earth with the light and heat necessary to support life





View called "Earthrise" that greeted *Apollo 8* astronauts as their spacecraft emerged from behind the Moon in December 1968. This classic image let people see Earth differently than ever before.



This image taken from *Apollo 17* in December 1972 is perhaps the first to be called "The Blue Marble." The dark blue ocean and swirling cloud patterns remind us of the importance of the oceans and atmosphere.

Earth system

- As we look closely at our planet from space, it becomes apparent that Earth is much more than rock and soil.
- In fact, among the most conspicuous features in both views of Earth in the previous figures are swirling clouds suspended above the surface of the vast global ocean.
- These features emphasize the importance of water on our planet.

Earth system

- The closer view of Earth from space helps us appreciate why the physical environment is traditionally divided into three major parts: the water portion of our planet, the hydrosphere; Earth's gaseous envelope, the atmosphere; and, of course, the solid Earth, or geosphere.
- It needs to be emphasized that our environment is highly integrated and not dominated by rock, water, or air alone. Rather, it is characterized by continuous interactions as air comes in contact with rock, rock with water, and water with air.

Earth system

- Moreover, the biosphere, which is the totality of all plant and animal life on our planet, interacts with each of the three physical realms and is an equally integral part of the planet. Thus, Earth can be thought of as consisting of four major spheres: the hydrosphere, atmosphere, geosphere, and biosphere

Earth system

- The interactions among Earth's spheres are countless. Next figure provides us with one easy-to-visualize example. The shoreline is an obvious meeting place for rock, water, and air. In this scene, ocean waves created by the drag of air moving across the water are breaking against the rocky shore

Earth system



Figure 1.11
Interactions among Earth's spheres The shoreline is one obvious interface—a common boundary where different parts of a system interact. In this scene at California's Montano del Oro State Park, ocean waves (hydrosphere) that were created by the force of moving air (atmosphere) break against a rocky shore (geosphere). The force of the water can be powerful, and the erosional work that is accomplished can be great. (Photo by Michael Collier)

- Hydrosphere: Earth is sometimes called the blue planet. Water, more than anything else, makes Earth unique.
- The hydrosphere is a dynamic mass of water that is continually on the move, evaporating from the oceans to the atmosphere, precipitating to the land, and running back to the ocean again.
- The global ocean is certainly the most prominent feature of the hydrosphere, blanketing nearly 71 % of Earth's surface to an average depth of about 3800 meters and accounts for about 97 % of Earth's water.

- However, the hydrosphere also includes the freshwater found underground and in streams, lakes, and glaciers. Moreover, water is an important component of all living things.
- Although these latter sources constitute just a tiny fraction of the total, they are much more important than their meager percentages indicates. In addition to providing the freshwater that is so vital to life on land, streams, glaciers, and groundwater are responsible for sculpting and creating many of our planet's varied landforms

Earth system

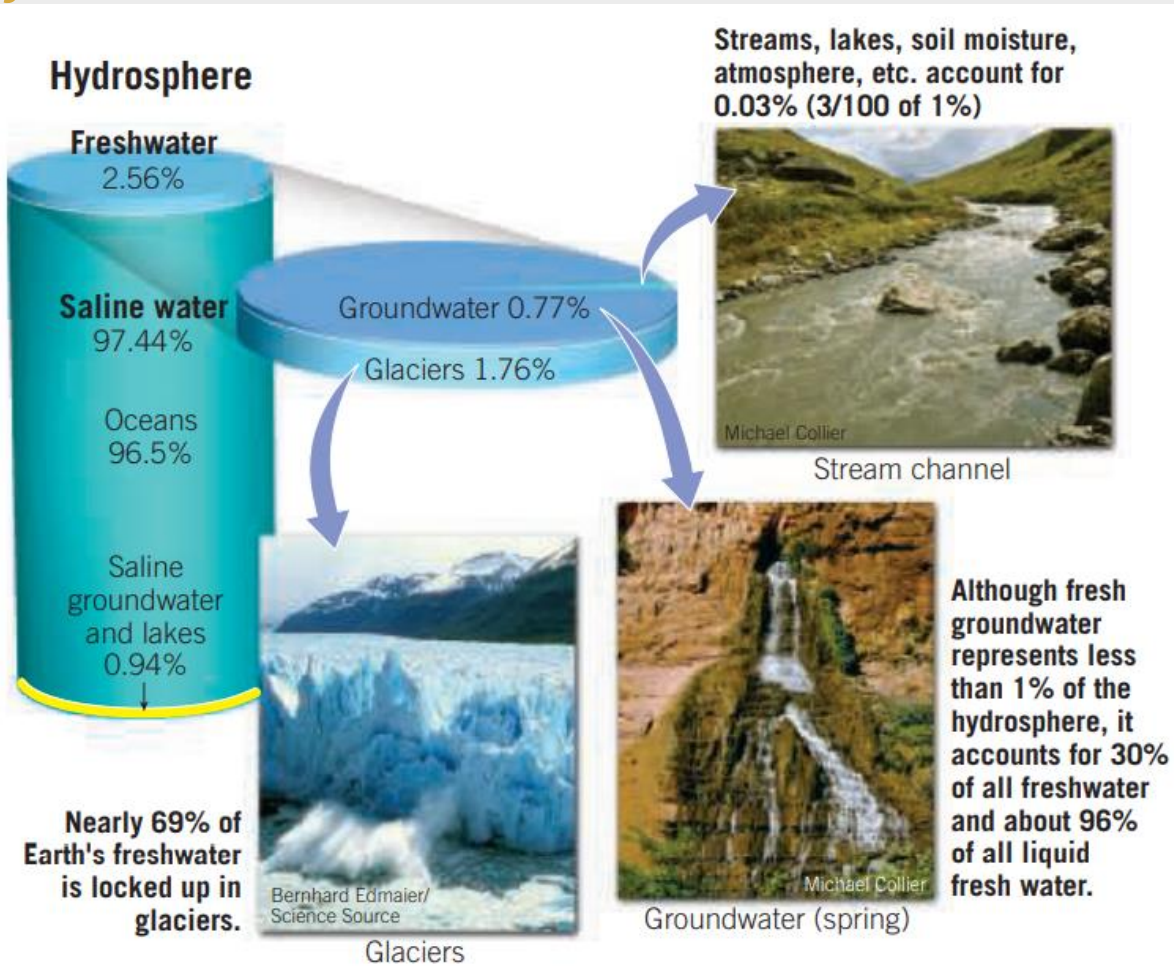


Figure 1.12
The water planet Distribution of water in the hydrosphere.

Earth system

- Atmosphere: Earth is surrounded by a life-giving gaseous envelope called the atmosphere.
- When we watch a high-flying jet plane cross the sky, it seems that the atmosphere extends upward for a great distance.
- However, when compared to the thickness (radius) of the solid Earth (about 6400 kilometers), the atmosphere is a very shallow layer.
- Despite its modest dimensions, this thin blanket of air is an integral part of the planet. It not only provides the air we breathe but also protects us from the Sun's intense heat and dangerous ultraviolet radiation.

- The energy exchanges that continually occur between the atmosphere and Earth's surface and between the atmosphere and space produce the effects we call weather and climate. Climate has a strong influence on the nature and intensity of Earth's external processes. When climate changes, these processes respond.
- If, like the Moon, Earth had no atmosphere, our planet would be lifeless, and many of the processes and interactions that make the surface such a dynamic place could not operate.
- Without weathering and erosion, the face of our planet might more closely resemble the lunar surface, which has not changed appreciably in nearly 3 billion years.

Earth system

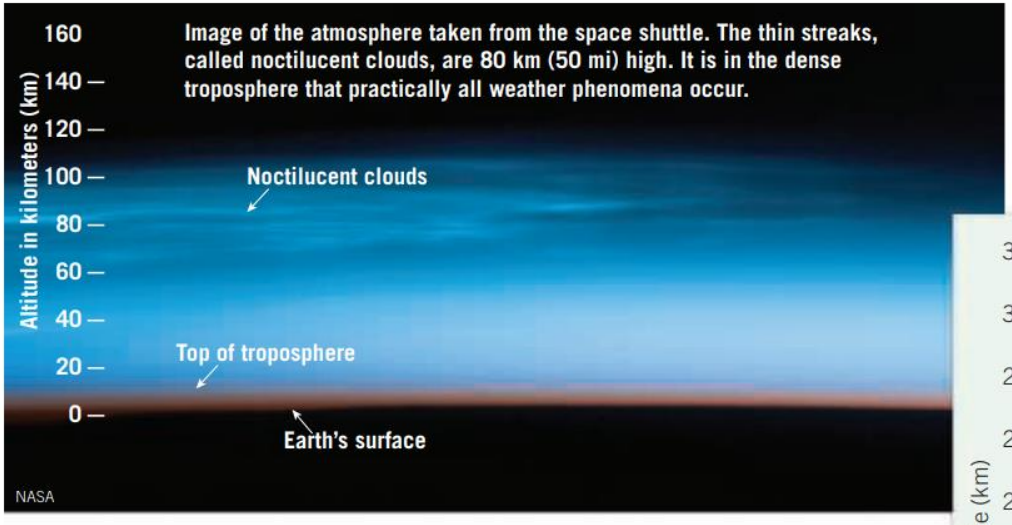
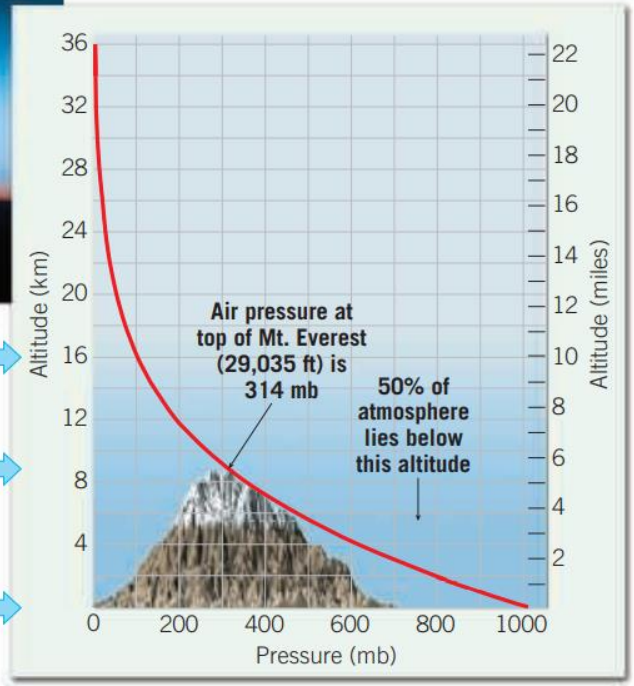


Figure 1.13
A shallow layer The atmosphere is an integral part of the planet. (NASA)

90% of the atmosphere is below 16 km (10 mi)

The air pressure atop Mt. Everest is about one-third that at sea level.

Average sea-level pressure is slightly more than 1000 millibars (about 14.7 lb./sq. in)



- Biosphere: includes all life on Earth. Ocean life is concentrated in the sunlit surface waters of the sea. Most life on land is also concentrated near the surface, with tree roots and burrowing animals tunneling a few meters underground and flying insects and birds reaching a kilometer or so into the atmosphere.
- A surprising variety of lifeforms are also adapted to extreme environments. For example, on the ocean floor, where pressures are extreme and no light penetrates, there are places where vents spew hot, mineral-rich fluids that support communities of exotic life-forms.

- On land, some bacteria thrive in rocks as deep as 4 km and in boiling hot springs.
- Moreover, air currents can carry microorganisms many kilometers into the atmosphere, But even when we consider these extremes, life still must be thought of as being confined to a narrow band very near Earth's surface.
- Plants and animals depend on the physical environment for the basics of life. However, organisms do not just respond to their physical environment. Through countless interactions, life-forms help maintain and alter the physical environment.

Earth system

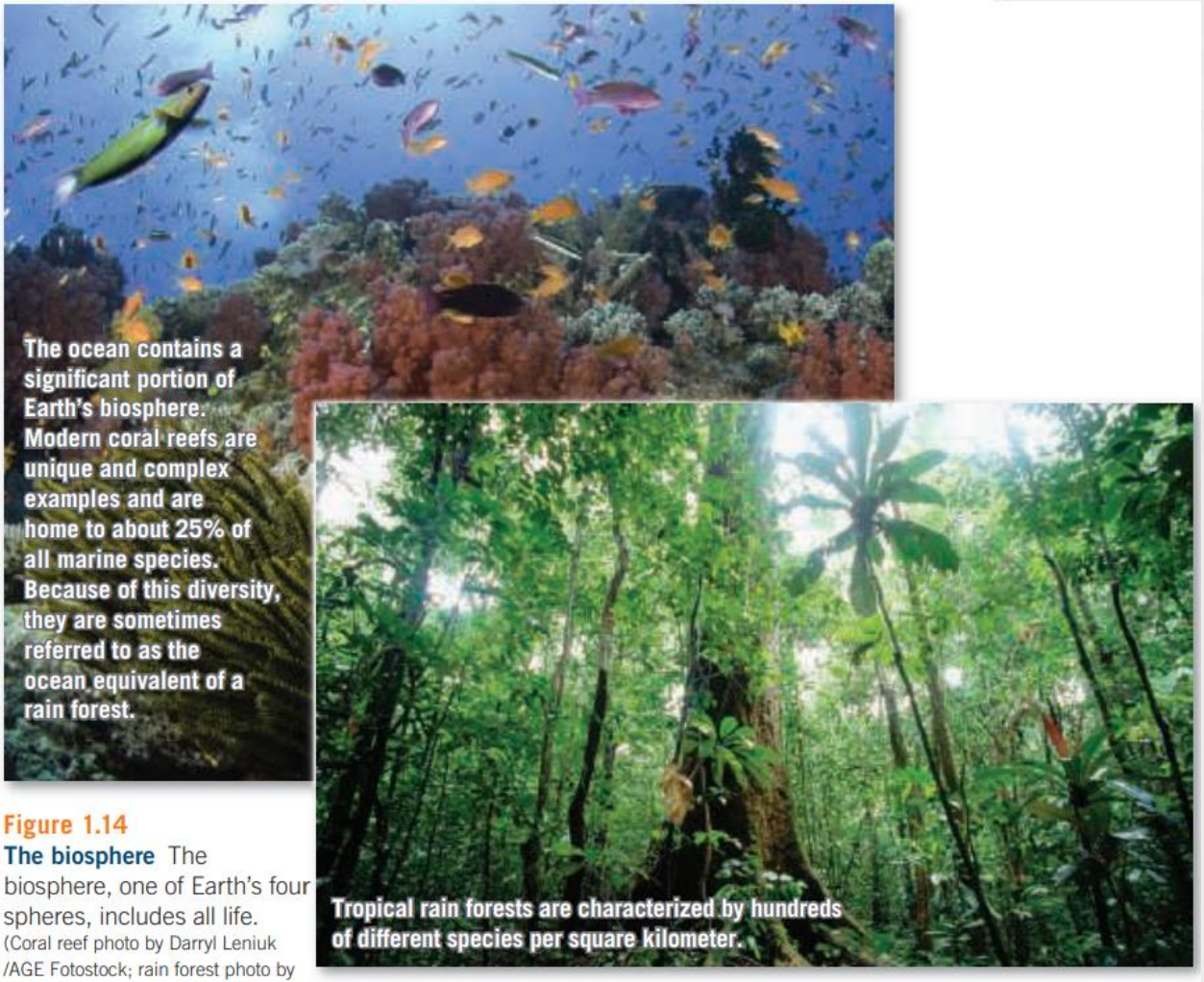


Figure 1.14
The biosphere The biosphere, one of Earth's four spheres, includes all life.
(Coral reef photo by Darryl Leniuk /AGE Fotostock; rain forest photo by AGE Fotostock/SuperStock)

- Geosphere: It beneath the atmosphere and the oceans is the solid Earth, or geosphere.
- The geosphere extends from the surface to the center of the planet, a depth of nearly 6400 km, making it by far the largest of Earth's four spheres.
- Much of our study of the solid Earth focuses on the more accessible surface features. Fortunately, many of these features represent the outward expressions of the dynamic behavior of Earth's interior.

- Examining the most prominent surface features and their global extent gives us clues to the dynamic processes that have shaped our planet.
- Soil, the thin veneer of material at Earth's surface that supports the growth of plants, may be thought of as part of all four spheres. The solid portion is a mixture of weathered rock debris (geosphere) and organic matter from decayed plant and animal life (biosphere). The decomposed and disintegrated rock debris is the product of weathering processes that require air (atmosphere) and water (hydrosphere). Air and water also occupy the open spaces between the solid particles.

- A simple example of the interactions among different parts of the Earth system occurs every winter and spring, as moisture evaporates from the Pacific Ocean and subsequently falls as rain in coastal hills and mountains, triggering destructive debris flows. The processes that move water from the hydrosphere to the atmosphere and then to the solid Earth have a profound impact on the plants and animals (including humans) that inhabit the affected regions



Figure 1.15
Deadly debris flow This image provides an example of interactions among different parts of the Earth system. Extraordinary rains triggered this debris flow (popularly called a mudslide) on March 22, 2014, near Oso, Washington. The mass of mud and debris blocked the North Fork of the Stillaguamish River and engulfed an area of about 2.6 square kilometers (1 square mile). Forty-three people perished. (Photo by Michael Collier)

Earth system science

- Scientists have recognized that in order to more fully understand our planet, they must learn how its individual components (land, water, air, and life-forms) are interconnected.
- This called Earth system science, aims to study Earth as a system composed of numerous interacting parts, or subsystems.
- Rather than look through the limited lens of only one of the traditional sciences—geology, atmospheric science, chemistry, biology, and so on—Earth system science attempts to integrate the knowledge of several academic fields

Earth system science

- Using an interdisciplinary approach, those engaged in Earth system science attempt to achieve the level of understanding necessary to comprehend and solve many of our global environmental problems.
- A system is a group of interacting, or interdependent, parts that form a complex whole. Most of us hear and use the term system frequently.
- A news report might inform us of an approaching weather system.
- Further, we know that Earth is just a small part of a larger system known as the solar system, which in turn is a subsystem of an even larger system, the Milky Way Galaxy.

Earth system

- The Earth system has a nearly endless array of subsystems in which matter is recycled over and over. One familiar subsystem is the hydrologic cycle. It represents the unending circulation of Earth's water among the hydrosphere, atmosphere, biosphere, and geosphere.
- Water enters the atmosphere through evaporation from Earth's surface and transpiration from plants. Water vapor condenses in the atmosphere to form clouds, which in turn produce precipitation that falls back to Earth's surface.

- Some of the rain that falls onto the land infiltrates to be taken up by plants or become groundwater, and some flows across the surface toward the ocean.
- Viewed over long time spans, the rocks of the geosphere are constantly forming, changing, and re-forming. The loop that involves the processes by which one rock changes to another is called the rock cycle.
- The cycles of the Earth system are not independent of one another; to the contrary, these cycles come in contact and interact in many places.

Earth system

- The parts of the Earth system are linked so that a change in one part can produce changes in any or all of the other parts.
- For example, when a volcano erupts, lava from Earth's interior may flow out at the surface and block a nearby valley. This new obstruction influences the region's drainage system by creating a lake or causing streams to change course.
- The large quantities of volcanic ash and gases that can be emitted during an eruption might be blown high into the atmosphere and influence the amount of solar energy that can reach Earth's surface. The result could be a drop in air temperatures over the entire hemisphere.

- Where the surface is covered by lava flows or a thick layer of volcanic ash, existing soils are buried. This causes soil-forming processes to begin anew to transform the new surface material into soil
- The soil that eventually forms will reflect the interactions among many parts of the Earth system—the volcanic parent material, the climate, and the impact of biological activity.
- Of course, there would also be significant changes in the biosphere. Some organisms and their habitats would be eliminated by the lava and ash, whereas new settings for life, such as a lake formed by a lava dam, would be created.

- Where the surface is covered by lava flows or a thick layer of volcanic ash, The potential climate change could also impact sensitive life-forms.
- The Earth system is characterized by processes that vary on spatial scales from fractions of millimeters to thousands of kilometers. Time scales for Earth's processes range from seconds to billions of years. As we learn about Earth, it becomes increasingly clear that despite significant separations in distance or time, many processes are connected, and a change in one component can influence the entire system.

Earth system

- The Earth system is powered by energy from two sources. The Sun drives external processes that occur in the atmosphere, in the hydrosphere, and at Earth's surface. Weather and climate, ocean circulation, and erosional processes are driven by energy from the Sun.
- Earth's interior is the second source of energy. Heat remaining from when our planet formed and heat that is continuously generated by radioactive decay power the internal processes that produce volcanoes, earthquakes,
- and mountains.

Earth system

- Humans are part of the Earth system, a system in which the living and nonliving components are interconnected. Therefore, our actions produce changes in all the other parts. When we burn gasoline and coal, dispose of our wastes, and clear the land, we cause other parts of the system to respond, often in unforeseen ways.

02. Geosphere

Massive rocky material

- It breakdown to form soil – to grow plants
- 99% of carbon is founded in rocks and sediments



03. Atmosphere

- Nitrogen N2 78%
- Oxygen O2 21%
- Other gases
  - ❖ Greenhouse gases
    - Water vapour
    - Carbon dioxide
    - Methane
    - Nitrous oxide
    - Ozone



Aerosols
Shiny Aerosols ---------- Reflect lights
Dark Aerosols ---------- Absorb lights

04. Hydrosphere



Water

- Oceans 97%
- Glaciers and ice sheets 2%
- Water droplets

Dark Oceans ……Absorbs Lights

Ice and clouds ……Reflects Lights

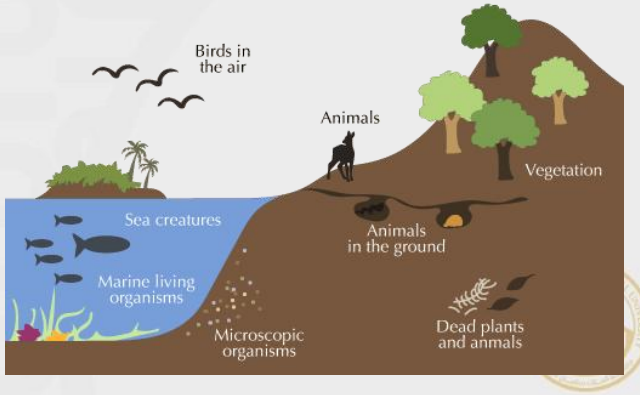
05. Biosphere

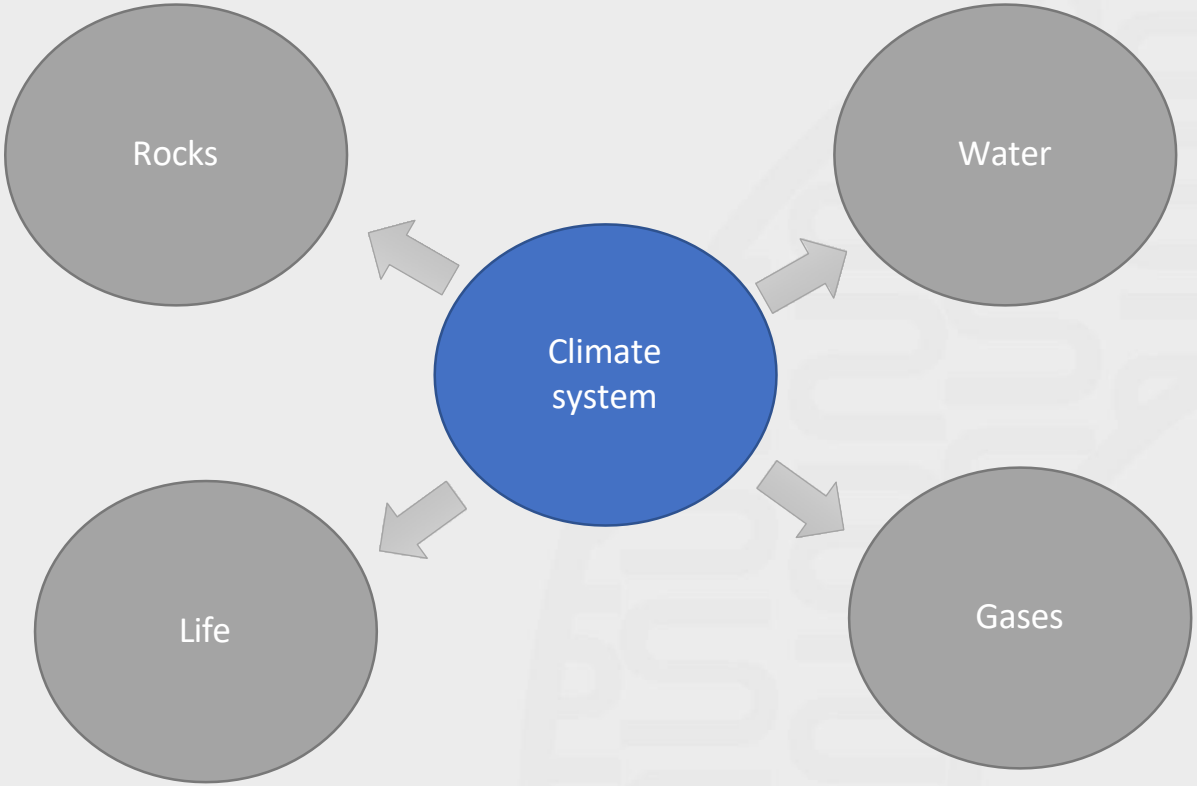
- Human Part of the biosphere – Deforestation
- Living things – exchange carbon and oxygen with atmosphere

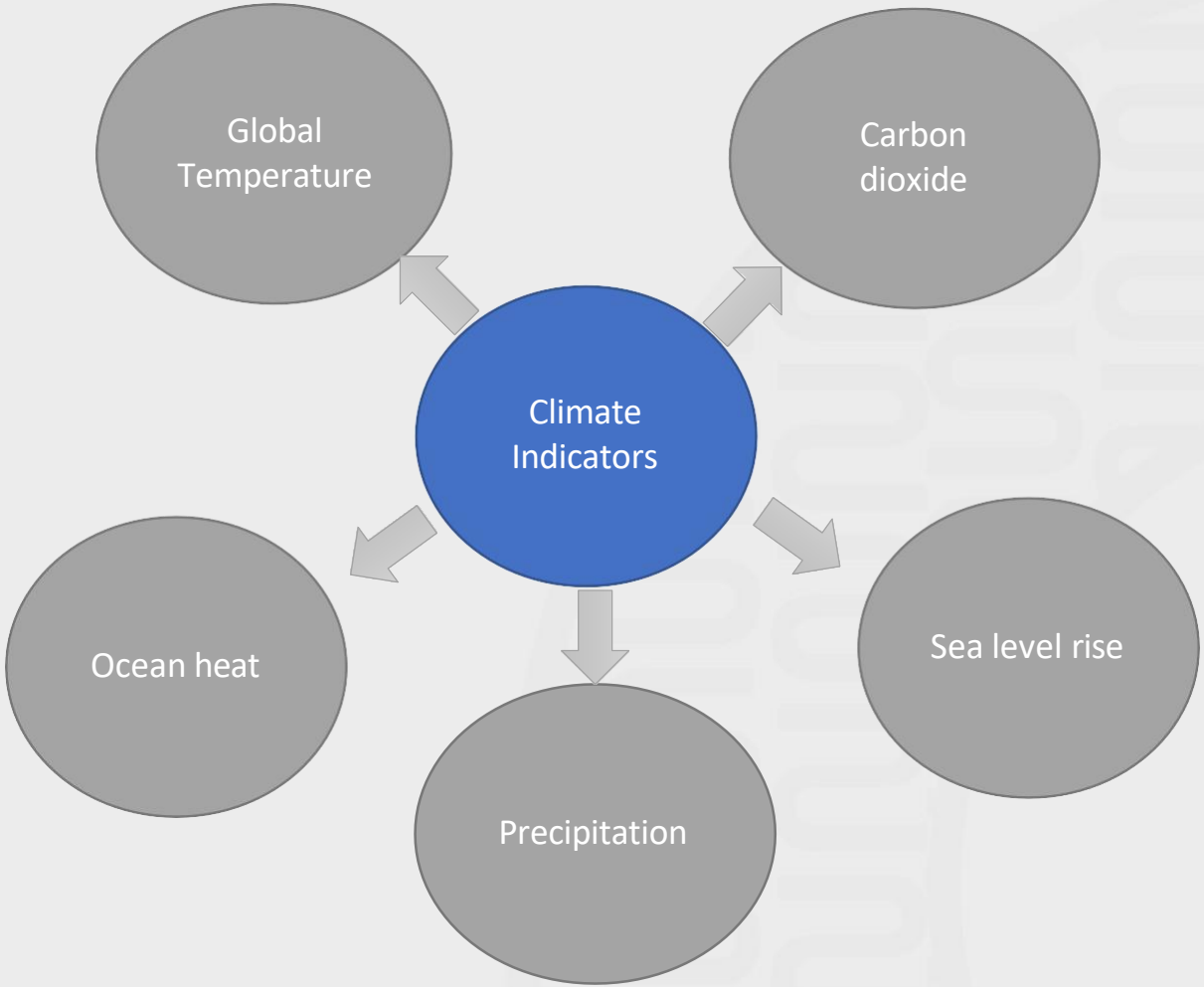
Vegetation ……absorbs Lights

Desert /grass lands ……reflects lights

In the oceans ……..life releases carbon




Rocks
Water
Climate system
Life
Gases


Global Temperature
Carbon dioxide
Climate Indicators
Ocean heat
Sea level rise
Precipitation

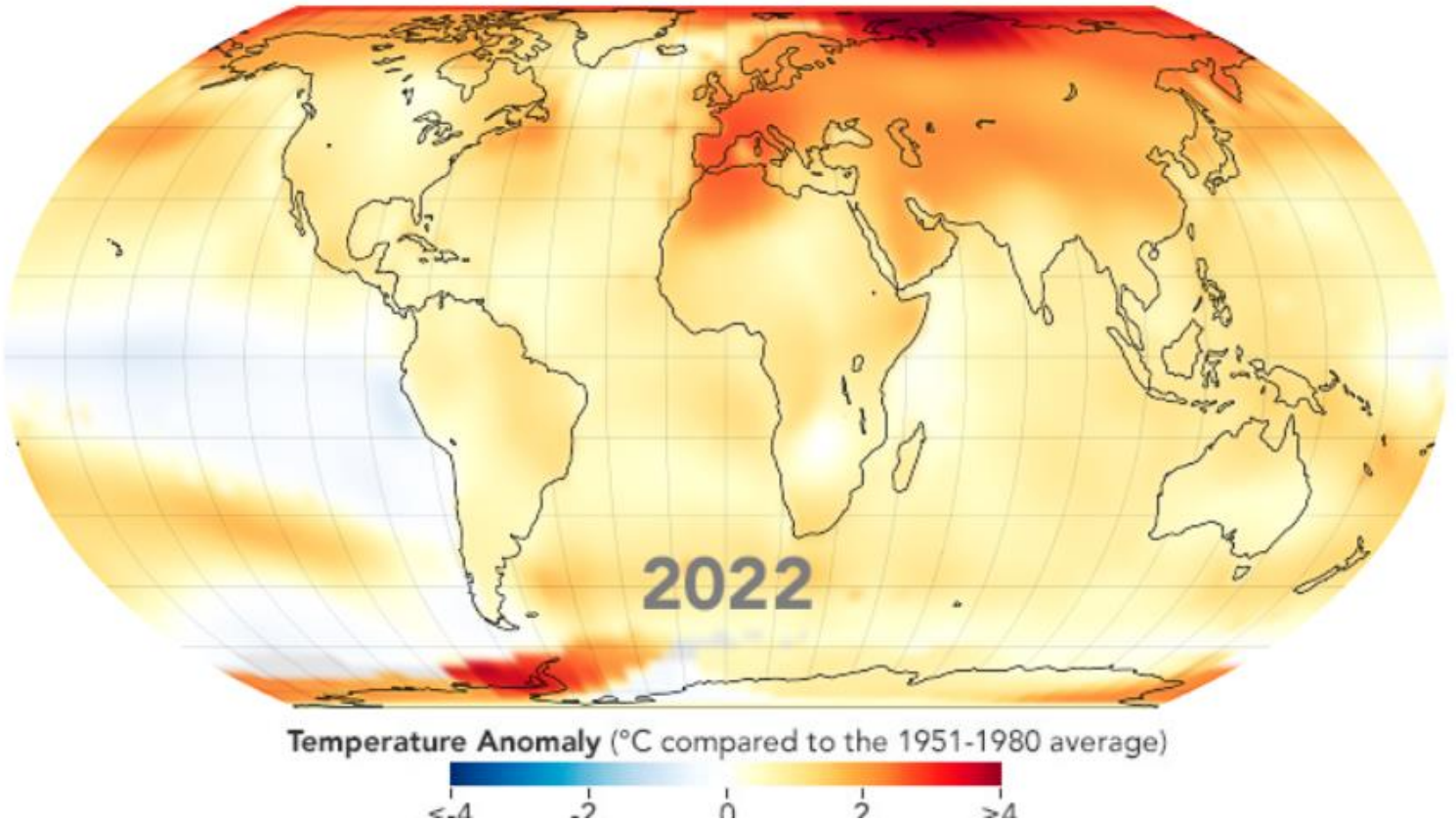
Global temperature

- Air temperatures on Earth have been rising since the Industrial Revolution. While natural variability plays some part, the majority of evidence indicates that human activities—particularly emissions of heat-trapping greenhouse gases—are mostly responsible for making our planet warmer.
- According to an ongoing temperature analysis, the average global temperature on Earth has increased by at least 1.1° Celsius since 1880. The majority of the warming has occurred since 1975, at a rate of roughly 0.15 to 0.20°C per decade (10 years).

Global temperature

- The maps show temperature anomalies in five-year increases since 1880. These are not absolute temperatures, but changes from the norm for each area. The data reflect how much warmer or cooler each region was compared to a base period of 1951-1980. (The global mean surface air temperature for that period was 14°C, with an uncertainty of several tenths of a degree.)
- The next image shows global temperature anomalies in 2022, which tied for the fifth warmest year on record. The past nine years have been the warmest years since modern recordkeeping began in 1880.

Global temperature



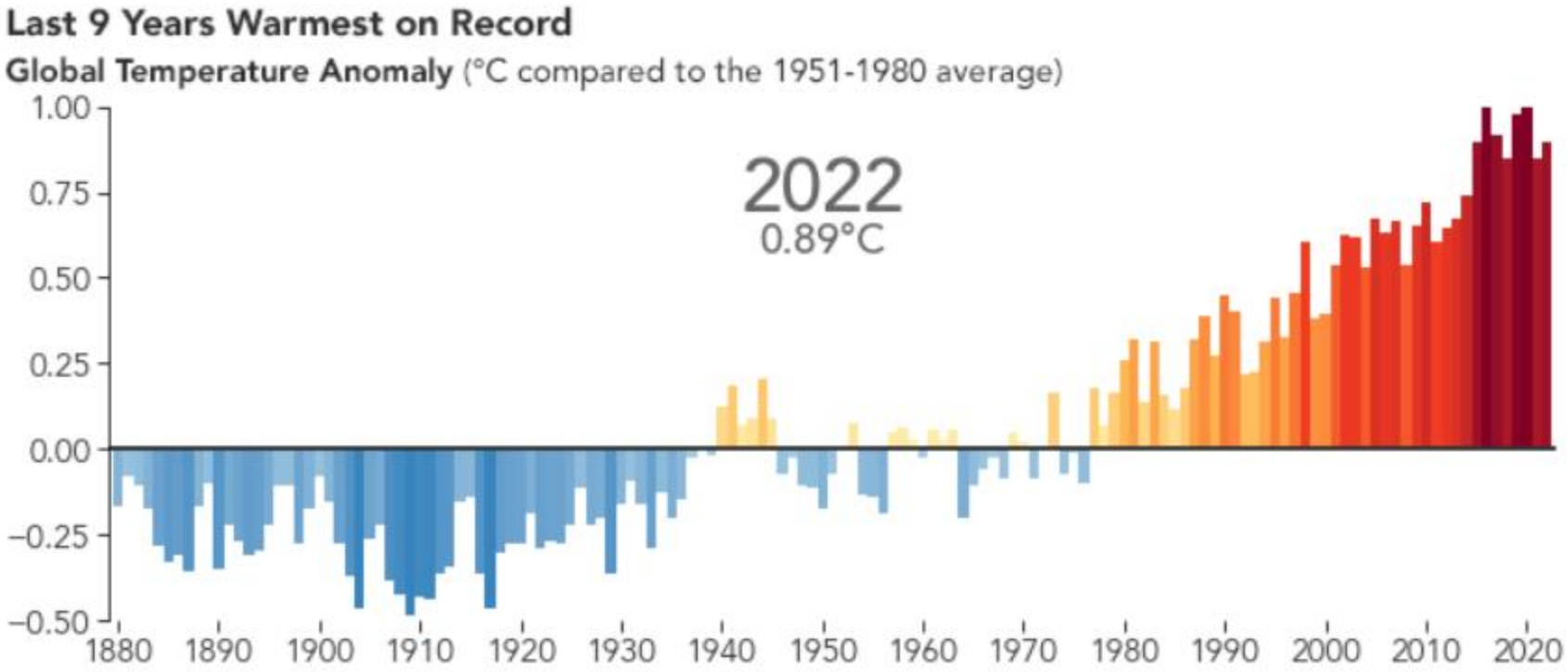
Global temperature

- As the maps show, global warming does not mean temperatures rise everywhere at every time by same rate. Temperatures might rise 5 degrees in one region and drop 2 degrees in another. For instance, exceptionally cold winters in one place might be balanced by extremely warm winters in another part of the world. Generally, warming is greater over land than over the oceans because water is slower to absorb and release heat. Warming may also differ substantially within specific land masses and ocean basins.

Global temperature

- In the last 150 years, the years from 1880 to 1939 tend to be cooler, then level off by the 1950s. Decades within the base period (1951-1980) do not appear particularly warm or cold because they are the standard against which other years are measured.
- The leveling off of temperatures in the middle of the 20th century can be explained by natural variability and by the cooling effects of aerosols generated by factories, power plants, and motor vehicles in the years of rapid economic growth after World War II. Fossil fuel use also increased after the war (5 % per year), boosting greenhouse gases. Cooling from aerosol pollution happened rapidly.

Global temperature



Global temperature

- In contrast, greenhouse gases accumulated slowly, but they remain in the atmosphere for a much longer time. The strong warming trend of the past four decades likely reflects a shift from balanced aerosol and greenhouse gas effects on the atmosphere to a predominance of greenhouse gas effects after aerosols were curbed by pollution controls.

Global temperature

- Why should we care about one or two degrees of global warming? After all, temperatures fluctuate by many degrees every day where we live due to cyclical events (night and day, summer and winter). But the global temperature mainly depends on Solar variability and the chemical composition of the atmosphere—particularly greenhouse gases.
- A 1° global change is significant because it takes a vast amount of heat to warm all of the oceans, the atmosphere, and the land masses by that much. In the past, a 1- to 2° drop was all it took to plunge the Earth into the Little Ice Age. A 5° drop was enough to inter a large part of North America under a towering mass of ice 20,000 years ago.

Carbon dioxide

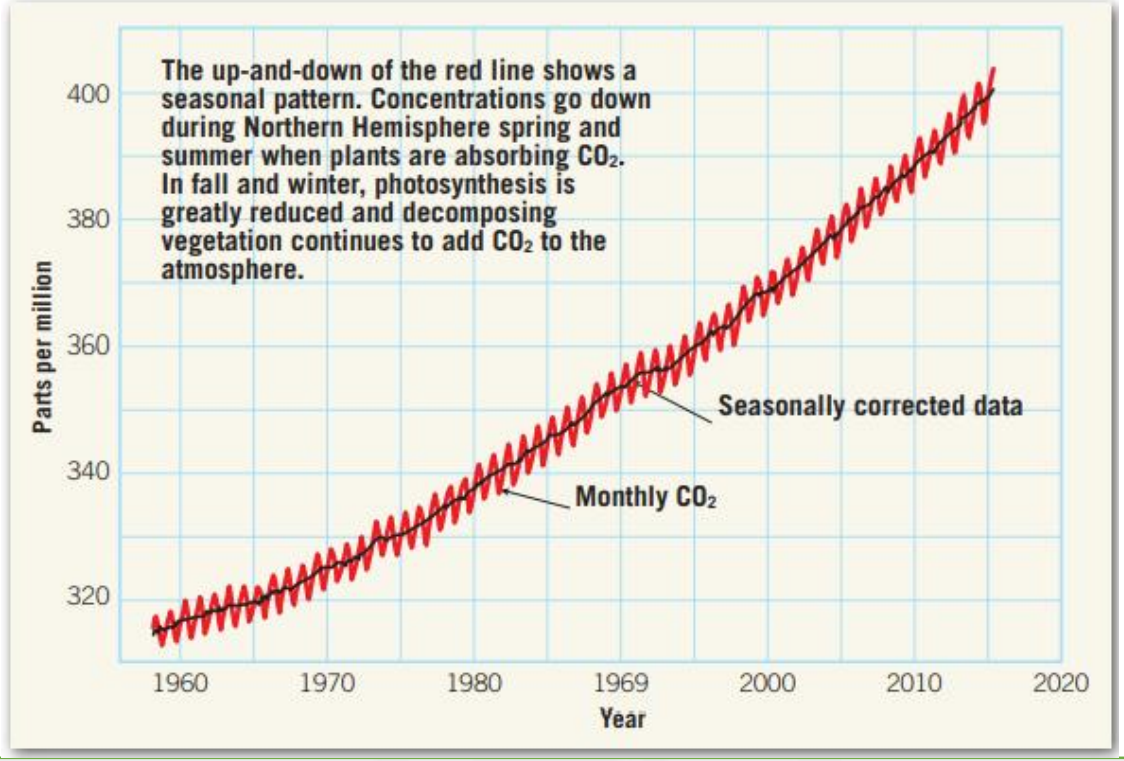
- Carbon dioxide (CO_2) represents only about 0.0400 percent (400 parts per million) of the gases that make up clean, dry air. Nevertheless, it is a very significant component meteorologically.
- CO_2 is influential because it is transparent to incoming short-wavelength solar radiation, but it is not transparent to some of the longer-wavelength, outgoing Earth radiation.
- The great industrialization of the past two centuries has been fueled—and still is fueled—by burning fossil fuels: coal, natural gas, and petroleum
-

Carbon dioxide

- Combustion of these fuels has added great quantities of carbon dioxide to the atmosphere. Next Figure shows changes in CO2 concentrations since 1958.
- The graph shows an annual seasonal cycle and a steady upward trend over the years. The up-and-down of the seasonal cycle is due to the vast land area of the Northern Hemisphere, which contains the majority of land-based vegetation. During spring and summer in the Northern Hemisphere, when plants are absorbing CO2 as part of photosynthesis, concentrations decrease.

Carbon dioxide

- The annual increase in atmospheric CO2 during the cold months occurs as vegetation dies and leaves fall and decompose, releasing CO2 back into the air.



Carbon dioxide

- The clearing of forests" deforestation" also contributes substantially because CO2 is released as vegetation is burned or decays. Deforestation is particularly pronounced in the tropical forests—including those in South America, Africa, Southeast Asia, and Indonesia—are disappearing.
- Some of the excess CO2 is taken up by plants or is dissolved in the ocean, but an estimated 45 percent remains in the atmosphere. The Next graph is a graphic record of changes in atmospheric CO2 extending back more than 400,000 years. Over this long span, natural fluctuations have varied from about 180 to 300 ppm.

Carbon dioxide

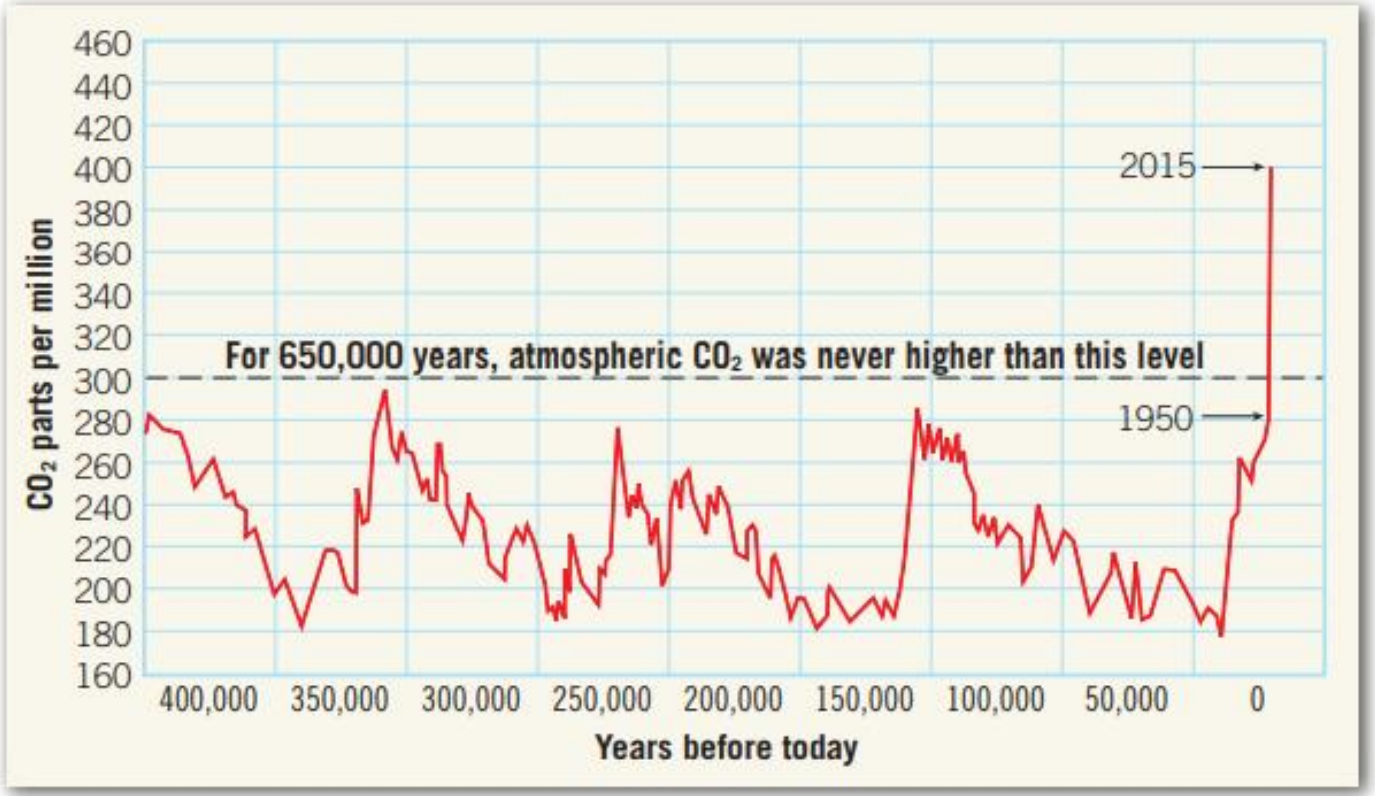


Figure 21.26
CO_2 concentrations over the past 400,000 years Most of these data come from the analysis of air bubbles trapped in ice cores. The record since 1958 comes from direct measurements at Mauna Loa Observatory, Hawaii. The rapid increase in CO_2 concentrations since the onset of the Industrial Revolution is obvious. (Based on NOAA)

Carbon dioxide



Figure 21.25
Tropical deforestation
Clearing the tropical rain forest is a serious environmental issue. In addition to causing a loss of biodiversity, deforestation is a significant source of carbon dioxide. Fires are frequently used to clear the land. This scene is in Brazil's Amazon basin. (Photo by Nigel Dickinson/Alamy)

Carbon dioxide

- As a result of human activities, the present CO2 level is about 30 %higher than its highest level over at least the past 400,000 years. The rapid increase in CO2 concentrations since the onset of industrialization is obvious. The annual rate at which atmospheric CO2 concentration is growing has been increasing over the past several decades.

Sea level rise

- A significant impact of human-induced global warming is a rise in sea level. As this occurs, coastal cities, wetlands, and low-lying islands could be threatened with more frequent flooding, increased shoreline erosion, and saltwater invasion into coastal rivers and aquifers.
- How is a warmer atmosphere related to a rise in sea level? One significant factor is thermal expansion. Higher air temperatures warm the adjacent upper layers of the ocean, which in turn causes the water to expand and sea level to rise. A second factor contributing to global sea-level rise is melting glaciers. With few exceptions, glaciers around the world have been retreating at unprecedented rates over the past century.

Ocean heat and acidity

- The human-induced increase in the amount of carbon dioxide in the atmosphere has some serious implications for ocean chemistry and for marine life.
- Recent studies show that about one-third of human-generated CO2 currently ends up in the oceans. The additional carbon dioxide lowers the ocean's pH, making seawater more acidic.
- When atmospheric CO_2 dissolves in seawater (H_2O), it forms carbonic acid (H_2CO_3). This lowers the ocean's pH and changes the balance of certain chemicals found naturally in seawater.

Ocean heat and acidity

- In fact, the oceans have already absorbed enough carbon dioxide for surface waters to have experienced a decrease of 0.1 pH units since preindustrial times, with an additional pH decrease likely in the future.
- This shift toward acidity and the resulting changes in ocean chemistry make it more difficult for certain marine creatures to build hard parts out of calcium carbonate.
- The decline in pH thus threatens a variety of calcite-secreting organisms as diverse as microbes and corals, which concerns marine scientists because of the potential consequences for other sea life that depend on the health and availability of these organisms.
- Ocean acidification in recent decades has been occurring 100 times faster than during past natural events over the last 55 million years.

Precipitation

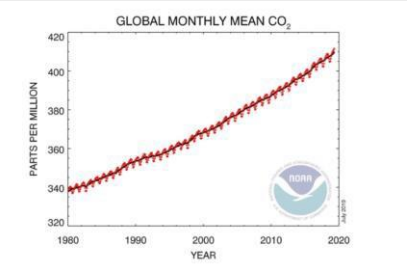
Precipitation is a form of water from the atmosphere. It is a term in meteorology, and includes rain, snow, sleet, ice pellets dew, frost, and hail. These form by condensation from atmospheric water vapor, and fall under gravity. Fog and mist are not precipitation but suspensions.

Greenhouse Gases

- In 2017, the global levels of CO2 concentration in the atmosphere passed 405 parts per million (ppm) and will not decrease for generations to come. This does not include other greenhouse gases or aerosols, which also affect the climate.

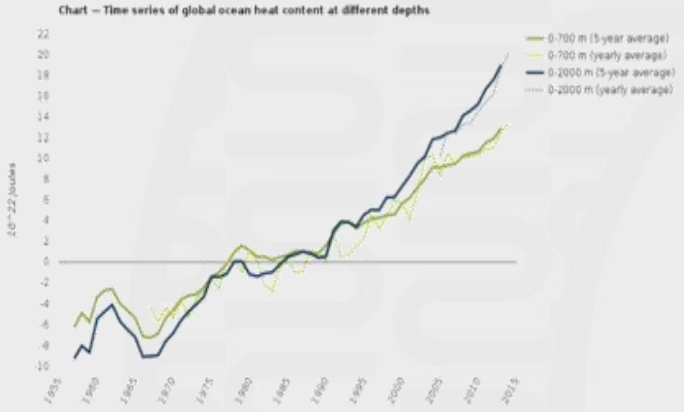
- The concentration of other greenhouse gases such as CH4 and N2O were also the highest on record in 2018.

- Together, land-use change and fossil CO2 emissions reached an estimated 41.5 ±3.0 billion tons of CO2 in 2018. You might find this also expressed in gig tons or GtCO2.



Ocean Heat

- **More than 90% of the energy trapped by greenhouse gases goes into the oceans. The global surface temperature has been rising slowly because our oceans have been absorbing most of the excess energy in the climate system.**



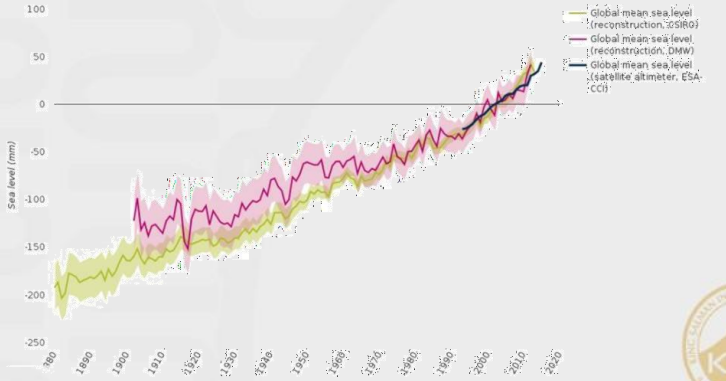
Sea Ice

- The extent and volume of Arctic sea ice have declined rapidly since global data became available, especially in summer.
- In each year between 2007 and 2018, Arctic summer sea ice extent was lower compared to any previous year since the introduction of satellites in 1979.
- The largest bodies of ice, the Greenland and Antarctic ice sheets, have been losing large amounts of ice at an increasing rate since 1992.

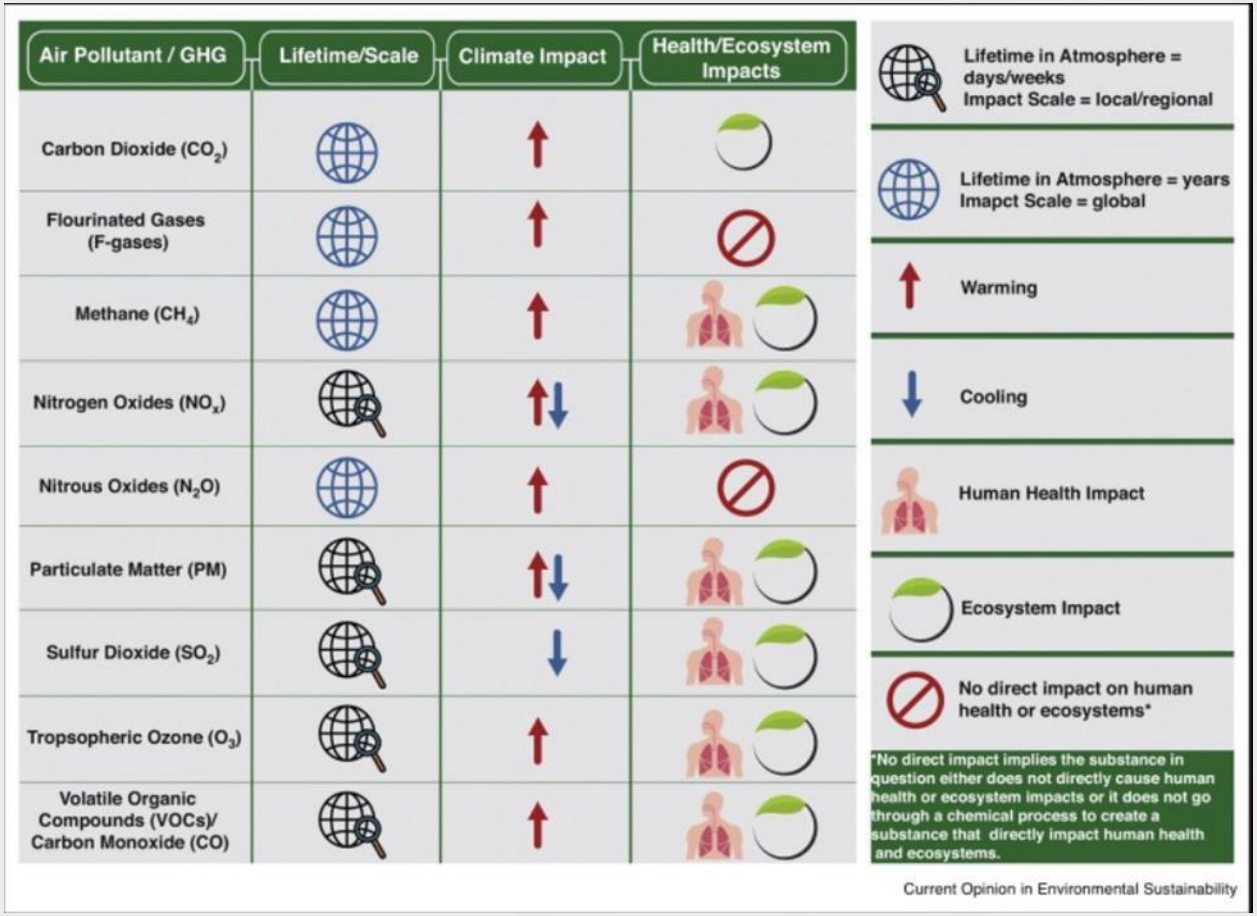


Sea Level

- Global mean sea level in 2016 was the highest yearly average since measurements started in the late 19th century.
- The rate of sea level rise since 1993, when satellite measurements became available, is around 3 mm/year, compared to 1.2 to 1.7 mm/year prior.
- In 2018, the total sea level change since 1993 was 78 mm.



Greenhouse Gases



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THANK YOU

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